

Strength of the Upper Bounds for the Edge-Weighted Maximum Clique Problem

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Abstract

We theoretically and computationally compare the strength of the three main upper bounds from the literature on the optimal value of the Edge-Weighted Maximum Clique Problem (EWMCP). We provide a set of instances for which the ratio between any of the three upper bounds and the optimal value of the EWMCP is unbounded, showing that none of them can give a performance guarantee. We further analyze the relative strength among the three upper bounds by determining, for every choice of a ratio between any two of them, the largest values it can attain and providing families of instances for which such values can be reached. Our results show that, for each pair of upper bounds, there exist appropriately chosen instances on which either bound is tighter than the other. Our theoretical analysis is complemented by extensive computational experiments on two benchmark datasets: the standard DIMACS instances and randomly generated instances, providing practical insights into the empirical strength of the upper bounds.

Keywords: Graph Theory · Combinatorial Optimization · Edge-Weighted Maximum Clique Problem

1 Introduction

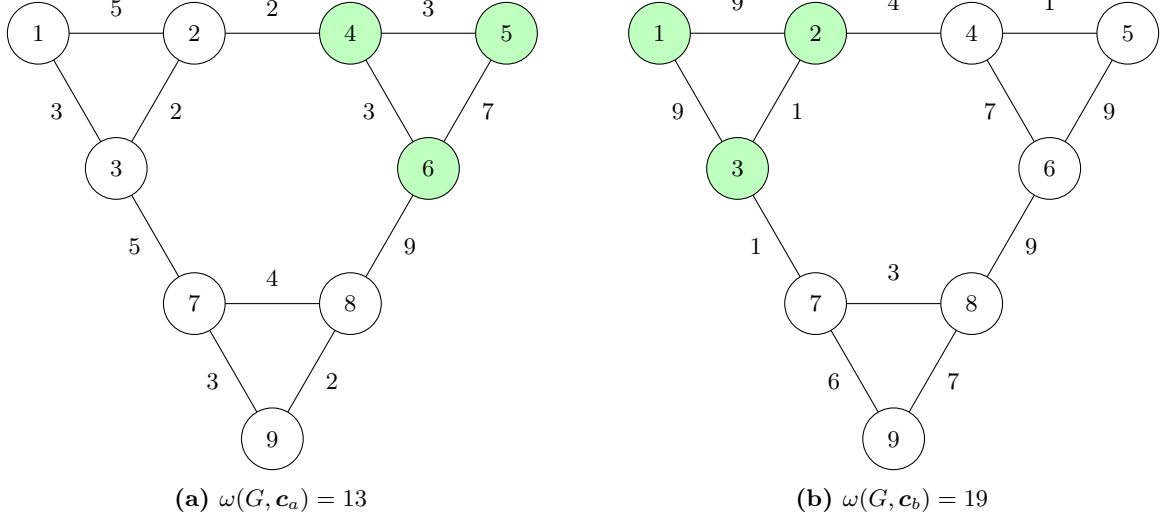
Let $G = (V, E)$ be a *simple undirected graph*, where V is the set of vertices, each labeled with a distinct natural number, and E is the set of edges connecting pairs of vertices. We denote an edge $\{u, v\} \in E$, with $u, v \in V$, by e . We assume that each edge $e \in E$ is associated with a nonnegative integer *edge-weight* $c_e \in \mathbb{Z}_{\geq 0}$. Given a clique $C \subseteq V$ of G , namely a subset of pairwise connected vertices, its *total edge-weight* is:

$$\sum_{e \in E[C]} c_e, \tag{1}$$

where $E[C] \subseteq E$ is the set of edges connecting the vertices in C . The *Edge-Weighted Maximum Clique Problem* (EWMCP) aims to determine a clique of the graph of maximum total edge-weight. The vector $\mathbf{c} \in \mathbb{Z}_{\geq 0}^{|E|}$, containing all edge-weights, is called the *edge-weight vector*. We denote an instance of the EWMCP by the pair $(G, \mathbf{c})$. We denote by $\omega(G, \mathbf{c})$ the optimal value of the EWMCP and call it the *edge-weighted clique number* of the graph. It is worth noticing that, since all edge-weights are nonnegative, there exists a maximum total edge-weight clique which is inclusion-wise maximal. Figure 1 introduces two EWMCP instances that are used throughout the remainder of the article and also illustrates their corresponding optimal solutions.

The EWMCP is an important generalization of the classical *Maximum Clique Problem* (MCP), which aims to determine a clique in a graph with the largest possible number of vertices. We

Figure 1: Two EWMCP instances with the same vertex set and edge set but different edge-weight vectors $\mathbf{c}_a$ and $\mathbf{c}_b$. The natural number labeling a vertex is represented inside the vertex, whereas the edge-weight of an edge is reported next to the edge. In Part (a), the clique $\{4, 5, 6\}$, highlighted in green, is a maximum total edge-weight clique for the first instance. The edge-weighted clique number of the graph $\omega(G, \mathbf{c}_a)$ is 13. In Part (b), the clique $\{1, 2, 3\}$, highlighted in green, is a maximum total edge-weight clique for the second instance. The edge-weighted clique number of the graph $\omega(G, \mathbf{c}_b)$ is 19.



refer the interested reader to San Segundo et al. (2023) for the state-of-the-art exact algorithm for the MCP. We denote by $\omega(G)$ the optimal value of the MCP and refer to it as the *clique number* of the graph. Since the MCP is strongly $\mathcal{NP}$ -hard, the same holds for the EWMCP (Gary and Johnson, 1979; Karp, 2010).

The EWMCP has several real-world applications across many domains, such as robotics, computer vision and materials science (Agapito et al., 2016; Ma and Latecki, 2012; San Segundo and Artieda, 2015; San Segundo et al., 2013). In many of these applications, edge weights encode the strength, similarity, or reliability of pairwise relationships, so that the goal is to identify a clique that is not only mutually compatible, but also maximizes the overall quality of the selected configuration. We refer the interested reader to other relevant weighted clique-related problems; see, for instance, Coniglio et al. (2021); Cornaz et al. (2017); Delle Donne et al. (2021); Furini et al. (2019, 2021); Mattia (2024); San Segundo et al. (2019).

Over the past decades, numerous exact approaches have been proposed in the literature to solve the EWMCP to proven optimality (Lei and Church, 2015; Prokopyev et al., 2009; San Segundo et al., 2019; Shimizu et al., 2019, 2020). The main state-of-the-art techniques are combinatorial branch-and-bound algorithms that efficiently explore the search space through an implicit enumeration (San Segundo et al., 2019; Shimizu et al., 2019, 2020; Hosseini et al., 2020). These algorithms rely on effective bounding techniques to limit the search space.

There are three main upper bounds for the optimal value of the EWMCP, which constitute the basis of the bounding procedures used in the exact algorithms proposed by San Segundo et al. (2019), Shimizu et al. (2020), and Hosseini et al. (2020). Recall that an *independent set* is a

subset of pairwise non-adjacent vertices, and that a *coloring* of the vertices is an assignment of colors to the vertices such that no two adjacent vertices receive the same color. Equivalently, it can be viewed as a partition of the vertex set into independent sets, each corresponding to the set of vertices receiving the same color. Since any feasible solution of the EWMCP is a clique, it can contain at most one vertex from each independent set in a coloring of the vertices. This observation plays a key role in the construction of all three upper bounds, although in different ways. The first upper bound is derived from the *Linear Programming* (LP) relaxation of an *Integer Linear Programming* (ILP) formulation of the EWMCP and exploits a coloring of the vertices to define the variables of the model. The second and third upper bounds are purely combinatorial: the former exploits a coloring of the vertices together with an ordering of the corresponding independent sets, while the latter relies on an upper bound on the clique number of the graph, which in practice can also be obtained from a coloring, since the number of colors provides an upper bound on the clique number. The first two upper bounds are formally defined in Section 2, while the third is defined in Section 5. To the best of our knowledge, these three upper bounds were developed independently and in parallel, and no comparison among them has been reported in the literature.

1.1 Performance comparison of exact algorithms for the EWMCP

From an algorithmic viewpoint, the three upper bounds are at the core of three state-of-the-art exact branch-and-bound algorithms for the EWMCP: BBEWC by San Segundo et al. (2019), MECQ+PLS by Shimizu et al. (2020), and the algorithm proposed by Hosseinian et al. (2020), which we denote by HFB. It is worth noticing that the computational experiments reported in the corresponding papers were performed on different machines and with different time limits, namely 2 hours for BBEWC and 3 hours for both MECQ+PLS and HFB. According to the results reported in the three papers, there are 48 DIMACS instances for which at least one of the three algorithms finds a certified optimal solution within its time limit. Among these, BBEWC solves 36 instances, MECQ+PLS solves 47, and HFB solves 38. Moreover, among the 48 instances, 9 are solved exclusively by MECQ+PLS and 1 exclusively by BBEWC, whereas no instances are solved only by HFB within the time limit.

1.2 Main contributions of the paper

The purpose of this article is to compare the upper bounds at a fundamental level: from a theoretical viewpoint, we investigate whether they can provide any performance guarantee with respect to $\omega(G, c)$ and whether there exist instances in which one bound is arbitrarily better than another, whereas from a computational viewpoint, we assess their behavior on the benchmark instances classically used in the literature for the EWMCP.

Our first theoretical result, presented in Section 3, shows in Sections 3.1 and 3.2 that the ratio between either of the first two upper bounds and the optimal value of the EWMCP is unbounded. Our second theoretical result, in Section 4, compares the first two upper bounds in both directions: in Section 4.1 we define a family of instances in which the second upper bound can be arbitrarily worse than the first, whereas in Section 4.2 we exhibit the opposite behavior.

These results, while ruling out the possibility of obtaining any relative performance guarantee with respect to the optimal value of the EWMCP, offer insight into the structural properties of the graphs that influence the strength of the first two upper bounds.

In Section 5, we introduce the third upper bound on $\omega(G, \mathbf{c})$. In Section 5.1, we compare it with the second upper bound: we prove that the second never exceeds twice the value of the third, that this factor is asymptotically tight, and that the third can be arbitrarily worse than the second. In Section 5.2, we show that the first and the third upper bounds are incomparable, as either can be arbitrarily worse than the other. Finally, in Section 5.3, we prove that the third upper bound also lacks a performance guarantee, as its ratio to $\omega(G, \mathbf{c})$ can be unbounded.

On the computational side, in Section 6 we report extensive results on two sets of instances to empirically assess the relative strength of the three upper bounds. After introducing the benchmark datasets in Section 6.1, we first analyze in Section 6.2 the sensitivity of UB_2 to the ordering of the independent sets in the coloring, showing that sorting them in non-decreasing order of cardinality consistently yields the best overall performance and effectively strengthens the practical behavior of this bound. We then report the main computational results on the classical DIMACS benchmark set in Section 6.3 and on randomly generated instances in Section 6.4, showing that the upper bound by Shimizu et al. (2020) dominates, in practice, those by San Segundo et al. (2019) and Hosseinian et al. (2020). Next, in Section 6.5 we evaluate the three bounds within a simulated branch-and-bound framework, measuring the tree depth at which each bound first drops to or below the optimal value, thereby allowing backtracking. Finally, in Section 6.6 we compare the first upper bound with other LP-based upper bounds on $\omega(G, \mathbf{c})$ derived from ILP formulations for the EWMCP proposed in the literature. In Section 7, we draw the main conclusions, highlighting the structural properties of graphs that favor one upper bound over the others, and outlining directions for future research.

2 The first two upper bounds on $\omega(G, \mathbf{c})$

In this section, we define the first two upper bounds on $\omega(G, \mathbf{c})$ that are analyzed and compared in this article. For completeness, we also include a proof of validity for each of the two upper bounds, following the derivations from the original works in which they were defined.

To this end, we first introduce the necessary notation. Let $I \subseteq V$ be a subset of pairwise non-adjacent vertices, called an *independent set*. Let $\mathcal{I}$ denote the collection of all independent sets of the graph. In addition, let $\mathcal{C} \subseteq \mathcal{I}$ be a partition of the vertex set V into disjoint, non-empty independent sets. We call such a partition a *coloring* of the vertices of the graph. We interpret the data of a coloring as an assignment of colors to the vertices such that any pair of connected vertices receives a different color.

2.1 The first upper bound

Following San Segundo et al. (2019), we describe in this section the derivation of the first upper bound on $\omega(G, \mathbf{c})$, which is based on a LP relaxation of an ILP formulation of the EWMCP,

with the constraints restricted to a given coloring.

For every $u \in V$, let x_u be a binary variable that takes value 1 if and only if vertex u is selected in the clique. For every edge $e \in E$, let y_e be a binary variable that takes value 1 if and only if both its endpoints u and v are selected in the clique. Using these binary variables, an ILP formulation for the EWMCP reads as follows:

$$\max \quad \sum_{e \in E} c_e y_e \quad (2a)$$

$$y_e \leq x_u, \quad e \in E, \quad (2b)$$

$$y_e \leq x_v, \quad e \in E, \quad (2c)$$

$$y_e \geq x_u + x_v - 1, \quad e \in E, \quad (2d)$$

$$\sum_{u \in I} x_u \leq 1, \quad I \in \mathcal{I}, \quad (2e)$$

$$x_u \in \{0, 1\}, \quad u \in V, \quad (2f)$$

$$y_e \in \{0, 1\}, \quad e \in E. \quad (2g)$$

The objective function (2a) of Model (2) maximizes the total edge-weight of the selected clique. Constraints (2b), (2c), and (2d) linearize the following logical implication: an edge $e \in E$ is part of the selected clique if and only if both its endpoints $u, v \in V$ are part of the selected clique. Since all edge-weights are nonnegative, Constraints (2d) can be dropped without affecting the set of optimal solutions of the model. For this reason, they are omitted in the remainder of this article. Finally, Constraints (2e) impose that, for every independent set of the graph, at most one vertex can belong to the selected clique. This ensures that the feasible solutions of Model (2) correspond to cliques in the graph. The optimal value of Model (2) is the edge-weighted clique number $\omega(G, \mathbf{c})$. It is worth noting that, without loss of generality, the binary constraints on the $\mathbf{y} \in \{0, 1\}^{|E|}$ variables can be dropped, since they are implicitly enforced by the binary nature of the $\mathbf{x} \in \{0, 1\}^{|V|}$ variables and the structure of the objective function. Therefore, the $\mathbf{y}$ variables can be considered as continuous free variables without affecting the correctness of the model.

By imposing the last set of constraints of Model (2) only on a subset $\tilde{\mathcal{I}}$ of $\mathcal{I}$, where each vertex $u \in V$ is contained in at least one independent set of $\tilde{\mathcal{I}}$, and by relaxing the integrality constraints on the variables $\mathbf{x}$ with nonnegativity constraints, we obtain a LP relaxation of the ILP Model (2).

To derive the dual of this LP relaxation, we introduce, for each edge $e \in E$, two dual variables ϱ_{eu} and ϱ_{ev} associated with the Constraints (2b) and (2c), respectively. In addition, for each $I \in \tilde{\mathcal{I}}$, we define a dual variable π_I corresponding to the associated Constraint (2e). The dual

of the LP relaxation described above reads as follows:

$$\text{UB}_1(G, \mathbf{c}, \tilde{\mathcal{J}}) := \min \sum_{I \in \tilde{\mathcal{J}}} \pi_I \quad (3a)$$

$$\varrho_{eu} + \varrho_{ev} = c_e, \quad e \in E, \quad (3b)$$

$$\sum_{I \in \tilde{\mathcal{J}}: u \in I} \pi_I \geq \sum_{e \in \delta(u)} \varrho_{eu}, \quad u \in V, \quad (3c)$$

$$\pi_I \geq 0, \quad I \in \tilde{\mathcal{J}}, \quad (3d)$$

$$\varrho_{ev} \geq 0, \quad e \in E, \quad (3e)$$

$$\varrho_{eu} \geq 0, \quad e \in E, \quad (3f)$$

where $\delta(u) \subseteq E$ is the set of all edges incident to vertex $u \in V$. By LP duality, the optimal value of Model (3) provides an upper bound on $\omega(G, \mathbf{c})$.

It is worth noting that the values of the variables ϱ , in any optimal solution of Model (3), can be interpreted as distributing the weight of each edge among its two endpoints. This interpretation is then used in the remainder of the article to illustrate the examples, and to make a comparison with the second upper bound. In more detail, for each edge $e \in E$, the optimal values of the variables ϱ_{eu} and ϱ_{ev} represent the portions of the edge-weight c_e assigned to the endpoints u and v , respectively. Moreover, for every $u \in V$, the quantity

$$\gamma_u := \sum_{e \in \delta(u)} \varrho_{eu} \quad (4)$$

can be interpreted as the total weight assigned to vertex u .

When $\tilde{\mathcal{J}} = \mathcal{C}$, as in San Segundo et al. (2019), Constraints (3c) can be written as

$$\pi_I \geq \sum_{e \in \delta(u)} \varrho_{eu} \quad I \in \mathcal{C}, u \in I,$$

and imply that, for each $I \in \mathcal{C}$, the variable π_I must be at least as large as the weight γ_u for every $u \in I$. In this interpretation, each variable π_I represents then the weight of the independent set I , and the objective function (3a) computes the total weight across all independent sets in the coloring $\mathcal{C}$. Then we have:

$$\text{UB}_1(G, \mathbf{c}, \mathcal{C}) = \sum_{I \in \mathcal{C}} \max \{ \gamma_u : u \in I \}. \quad (5)$$

Note that the upper bound employed in the branch-and-bound algorithm of San Segundo et al. (2019) corresponds to a heuristic solution of Model (3), where $\tilde{\mathcal{J}}$ is a suitably defined vertex coloring $\mathcal{C}$, and it is computed on subgraphs generated during the branching process. The

proposed heuristic procedure is designed to reduce the computational effort while computing good-quality upper bounds. In contrast, in this paper we focus on evaluating the strength of the upper bound itself. To this end, we consider $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$, which corresponds to the best possible upper bound within this family.

2.2 The second upper bound

Following Shimizu et al. (2020), we describe in this section the derivation of the second upper bound on $\omega(G, \mathbf{c})$, obtained by exploiting the structure of feasible EWMCP solutions and a coloring of the vertices of the graph.

Given a coloring $\mathcal{C} \subseteq \mathcal{I}$, let $\prec$ be a total order on $\mathcal{C}$, and let $I(u) \in \mathcal{C}$ be the unique independent set of $\mathcal{C}$ containing vertex $u \in V$, i.e., $u \in I(u)$.

In this and all subsequent occurrences, in an expression of the form $\max\{c_e : e \in \delta(u), v \in I\}$, v denotes the other endpoint of $e = \{u, v\}$, and the maximum is defined as 0 if no such edge exists.

For every $u \in V$, we define the quantity

$$\sigma_u := \sum_{I \in \mathcal{C} : I \prec I(u)} \max\{c_e : e \in \delta(u), v \in I\}, \quad (6)$$

which can be interpreted as the total weight assigned to vertex u . This weight corresponds to the sum, over all independent sets that precede the one containing u in the ordering, of the maximum edge-weight connecting u to a vertex in each of those independent sets.

The second upper bound studied in this article, denoted by $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$, is then defined as:

$$\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec) := \sum_{I \in \mathcal{C}} \max\{\sigma_u : u \in I\}. \quad (7)$$

To verify that this is indeed an upper bound for $\omega(G, \mathbf{c})$, consider any clique C in G . By construction, each vertex in the clique C belongs to a different independent set in the coloring $\mathcal{C}$. For every independent set $I \in \mathcal{C}$, let u_I be the unique vertex in C that belongs to I , if such a vertex exists; otherwise, let u_I be any vertex in I . We then obtain:

$$\sum_{e \in E[C]} c_e = \sum_{\substack{I \in \mathcal{C}: \\ C \cap I \neq \emptyset}} \sum_{\substack{J \in \mathcal{C}: \\ J \prec I, C \cap J \neq \emptyset}} c_{\{u_I, u_J\}} \quad (8a)$$

$$\leq \sum_{\substack{I \in \mathcal{C}: \\ C \cap I \neq \emptyset}} \max \left\{ \sum_{\substack{J \in \mathcal{C}: \\ J \prec I, C \cap J \neq \emptyset}} c_{\{u, u_J\}} : u \in I \right\} \quad (8b)$$

$$\leq \sum_{\substack{I \in \mathcal{C}: \\ C \cap I \neq \emptyset}} \max \left\{ \sum_{\substack{J \in \mathcal{C}: \\ J \prec I}} \max \{c_e : e \in \delta(u), v \in J\} : u \in I \right\} \quad (8c)$$

$$\leq \underbrace{\sum_{I \in \mathcal{C}} \max \{\sigma_u : u \in I\}}_{= \text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)}. \quad (8d)$$

The equality in (8a) rewrites the sum of the weights of the edges in the clique C by indexing over pairs of independent sets in $\mathcal{C}$ that contain vertices of C . The inequality in (8b) is obtained by maximizing the inner sum over all possible choices of a vertex $u \in I$. The inequality in (8c) follows by maximizing each term of the inner sum over all possible choices of a vertex $v \in J$ and by extending the sum to all independent sets $J \prec I$. Finally, the inequality in (8d) is obtained by extending the sum to all independent sets in $\mathcal{C}$. This derivation follows the proof of (Shimizu et al., 2020, Lemma 1).

Note that the upper bound $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ depends not only on a coloring, but also on the total order $\prec$ of the independent sets within the coloring. Using the same partition into independent sets, but varying the ordering of the independent sets, could result in different upper bounds. In Section 6.2 we empirically measure the impact of different orderings on the quality of this upper bound.

It is worth mentioning that Shimizu et al. (2020) also proposed an alternative upper bound, which can be interpreted as the value of a specific feasible solution to the dual formulation (3). The main idea is to assign the full weight of each edge $e \in E$ to the endpoint belonging to the independent set which precedes the other in $\mathcal{C}$ according to the total order $\prec$. Assuming without loss of generality that for each edge $e \in E$, we have $I(u) \prec I(v)$, this feasible solution is defined as follows:

$$\tilde{q}_{eu} := c_e, \quad \tilde{q}_{ev} := 0, \quad e \in E, \quad \tilde{\pi}_I := \max \left\{ \sum_{e \in \delta(u)} \tilde{q}_{eu} : u \in I \right\}, \quad I \in \mathcal{C}.$$

The upper bound obtained from this dual solution is, by construction, greater than or equal to the optimal value of the dual Model (3). Accordingly, the second upper bound proposed by

Shimizu et al. (2020) is dominated by $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$, and is therefore not considered in our study.

2.3 Examples of the upper bounds computation

In Figure 2, we report the values of the first two upper bounds for the graph shown in Figure 1 and its two different edge-weight vectors. The figure also includes all the values needed to compute the upper bounds, as described in Sections 2.1 and 2.2. These examples illustrate that there exist graphs, edge-weight vectors, and colorings for which $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ is smaller than $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ (see Parts 2a and 2c of Figure 2), and vice versa (see Parts 2b and 2d of Figure 2).

3 Theoretical comparison of the first two upper bounds with $\omega(G, \mathbf{c})$

In this section, we provide a sequence of EWMCP instances in which the ratio between either of the two upper bounds $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ and the edge-weighted clique number becomes arbitrarily large as the number of vertices in the graphs tends to infinity.

For every integer $n \geq 2$, let $G_n^1 = (V_n^1, E_n^1)$ be a graph with a vertex set containing $2n$ vertices, partitioned into two cliques $C_1, C_2 \subseteq V_n^1$, where

$$C_1 = \{1, 2, \dots, n\}, \quad C_2 = \{n+1, n+2, \dots, 2n\}.$$

The edge set of the graph includes all edges within the two cliques, each assigned an edge-weight of 1. Moreover, for each vertex $u \in C_1$, there is an edge $\{u, n+u\}$ with edge-weight

$$\bar{c} = \frac{n(n-1)}{2} - 1,$$

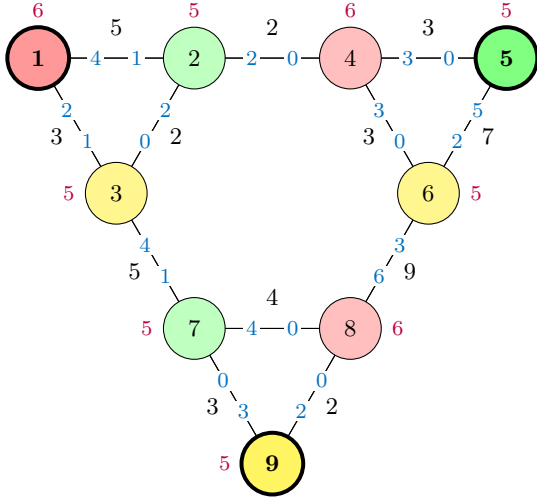
and, by construction, the vertex $n+u$ belongs to C_2 . These edge-weights define the edge-weight vector $\mathbf{c}_n^1$ for the graph G_n^1 . An illustration of G_n^1 is provided in Figure 3.

The inclusion-wise maximal cliques of G_n^1 are C_1, C_2 , with total edge-weight $\frac{n(n-1)}{2}$, and the n cliques of cardinality 2 given by the edges connecting C_1 and C_2 , with total edge-weight equal to $\bar{c}$. Hence, the edge-weighted clique number of the EWMCP instance $(G_n^1, \mathbf{c}_n^1)$ is

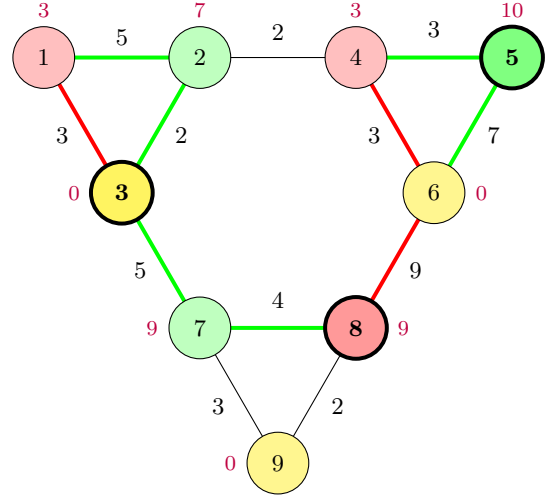
$$\omega(G_n^1, \mathbf{c}_n^1) = \frac{n(n-1)}{2}. \tag{9}$$

Since G_n^1 contains a clique of cardinality n , any coloring of its vertices requires at least n colors. It follows that for every coloring $\mathcal{C}$ of the vertices of G_n^1 , we have $|\mathcal{C}| \geq n$. Moreover, any independent set I in any coloring $\mathcal{C}$ contains at most one vertex from C_1 and at most one from C_2 , and therefore $|I| \leq 2$.

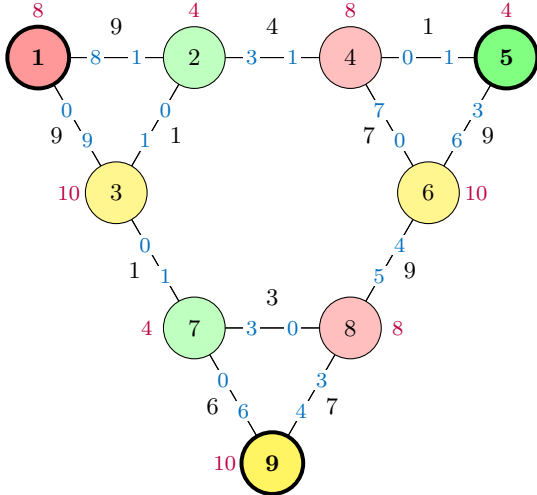
Figure 2: The values of the two upper bounds $UB_1(G, \mathbf{c}, \mathcal{C})$ and $UB_2(G, \mathbf{c}, \mathcal{C}, \prec)$ for the instances in Figure 1, with respect to the following coloring: $\{3, 6, 9\}$ (yellow), $\{1, 4, 8\}$ (red), $\{2, 5, 7\}$ (green). The total order is given by $\{3, 6, 9\} \prec \{1, 4, 8\} \prec \{2, 5, 7\}$. For each edge $e \in E$, the black label next to the edge shows its weight — in Parts 2a and 2b, this corresponds to the edge-weight vector $\mathbf{c}_a$ of Part 1a of Figure 1; in Parts 2c and 2d, to the edge-weight vector $\mathbf{c}_b$ of Part 1b of Figure 1. In Parts 2a and 2c, for each edge $e \in E$, the blue labels on the edge report the values of the variables ϱ_{eu} and ϱ_{ev} in an optimal solution of Model (3). For each vertex $u \in V$, the value γ_u (see the definition (4)) is shown in purple next to the vertex. For each color $I \in \mathcal{C}$, the vertex $u \in I$ with the maximum weight γ_u is highlighted in bold; this value corresponds to the value of the variable π_I in the same optimal solution of Model (3). In Parts 2b and 2d, for each vertex $u \in V$, the value σ_u (see the definition (6)) is shown in purple next to the vertex. Additionally, the edges contributing to σ_u are drawn in the same color as vertex u . For each color $I \in \mathcal{C}$, the vertex $u \in I$ with the maximum weight σ_u is highlighted in bold.



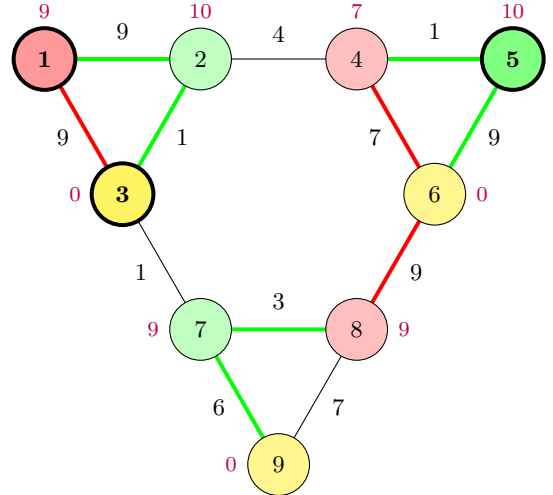
(a) $UB_1(G, \mathbf{c}_a, \mathcal{C}) = 16$



(b) $UB_2(G, \mathbf{c}_a, \mathcal{C}, \prec) = 19$

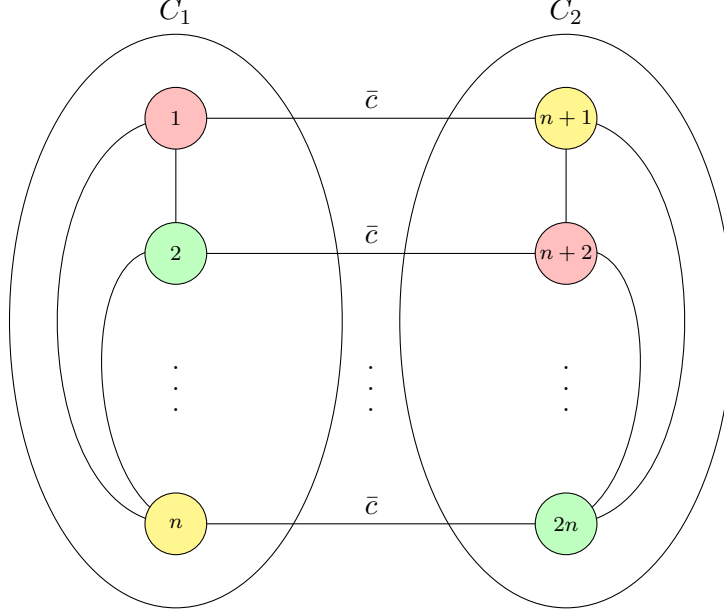


(c) $UB_1(G, \mathbf{c}_b, \mathcal{C}) = 22$



(d) $UB_2(G, \mathbf{c}_b, \mathcal{C}, \prec) = 19$

Figure 3: Representation of the graph G_n^1 , where the vertices are colored using the minimum number of colors. Each vertex is labeled with a natural number displayed inside the node. The two ellipses represent the cliques C_1 and C_2 . The edge-weight $\bar{c}$ of each edge connecting a vertex in C_1 to a vertex in C_2 is shown next to the edge. All edges within each clique have weight one; these values are omitted from the figure.



3.1 The first upper bound versus $\omega(G, \mathbf{c})$

The following proposition states that the ratio between the upper bound $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and edge-weighted clique number $\omega(G, \mathbf{c})$ can be arbitrarily large.

Proposition 3.1. *The sequence $\{(G_n^1, \mathbf{c}_n^1)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_1(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^1$ is any coloring of the vertices of G_n^1 .

Proof. Given any independent set $I \in \mathcal{C}_n^1$, by adding up the Constraints (3c) related to I , we obtain:

$$|I| \pi_I \geq \sum_{u \in I} \sum_{e \in \delta(u)} \varrho_{eu} \implies 2 \pi_I \geq \sum_{u \in I} \sum_{e \in \delta(u)} \varrho_{eu},$$

since $|I| \leq 2$. Moreover, adding up all these last inequalities for every independent set $I \in \mathcal{C}_n^1$, we then have:

$$2 \sum_{I \in \mathcal{C}_n^1} \pi_I \geq \sum_{e \in E} (\varrho_{eu} + \varrho_{ev}),$$

since any coloring is a partition of the vertex set. Then, according to Constraints (3b), the

following holds:

$$\underbrace{\sum_{I \in \mathcal{C}_n^1} \pi_I}_{= \text{UB}_1(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1)} \geq \frac{1}{2} \sum_{e \in E_n^1} c_e = \frac{1}{2} \sum_{e \in E_n^1[C_1]} c_e + \frac{1}{2} \sum_{e \in E_n^1[C_2]} c_e + \frac{1}{2} n \bar{c} = \underbrace{\frac{n(n-1)}{2}}_{= \omega(G_n^1, \mathbf{c}_n^1)} + \frac{1}{2} n \bar{c}.$$

Finally, it follows that:

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_1(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} \geq \lim_{n \rightarrow +\infty} 1 + \frac{n}{2} - \frac{1}{n-1} = +\infty.$$

□

3.2 The second upper bound versus $\omega(G, \mathbf{c})$

The following proposition states that the ratio between the upper bound $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ and edge-weighted clique number $\omega(G, \mathbf{c})$ can be arbitrarily large.

Proposition 3.2. *The sequence $\{(G_n^1, \mathbf{c}_n^1)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^1$ is any coloring of the vertices of G_n^1 and $\prec_n^1$ defines any total order of the independent sets in $\mathcal{C}_n^1$.

Proof. For any edge $\{w, n+w\} \in E_n^1$ with $w \in \{1, 2, \dots, n\}$, connecting a vertex in C_1 with a vertex in C_2 , its endpoints belong to distinct independent sets of $\mathcal{C}_n^1$. Let $V' \subseteq V_n^1$ be the subset of vertices $w \in C_1$ such that $I(w) \prec_n^1 I(n+w)$ and vertices $n+w \in C_2$ such that $I(n+w) \prec_n^1 I(w)$. Let $V'' := V_n^1 \setminus V'$. This implies that each vertex v in V' is connected to a vertex u in V'' with an edge of weight $\bar{c}$ and we have $I(v) \prec_n^1 I(u)$. Then, we have:

$$\sum_{v \in V_n^1} \sigma_v = \sum_{v \in V'} \sigma_v + \sum_{u \in V''} \sigma_u = \sum_{v \in V'} \sigma_v + \sum_{u \in V''} \sum_{J \in \mathcal{C}_n^1: J \prec_n^1 I(u)} \max\{c_e: e \in \delta(u), v \in J\}.$$

Since, for every $u \in V''$ there exists an edge of weight $\bar{c}$ belonging to the set $\{e \in \delta(u): v \in J, J \prec_n^1 I(u)\}$, and since $|V''| = n$, we have:

$$\sum_{v \in V_n^1} \sigma_v \geq n \bar{c}.$$

For every $I \in \mathcal{C}_n^1$ we have:

$$\max\{\sigma_u: u \in I\} \geq \frac{1}{2} \sum_{u \in I} \sigma_u,$$

since the right-hand side is at most as large as the average of the values in $\{\sigma_u: u \in I\}$. The

latter is true because $|I| \leq 2$ for every $I \in \mathcal{C}_n^1$. Adding up these inequalities for all $I \in \mathcal{C}_n^1$ yields

$$\underbrace{\sum_{I \in \mathcal{C}_n^1} \max\{\sigma_u : u \in I\}}_{= \text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1)} \geq \frac{1}{2} \sum_{v \in V_n^1} \sigma_v \geq \frac{n \bar{c}}{2} = \frac{n}{2} \left(\underbrace{\frac{n(n-1)}{2}}_{= \omega(G_n^1, \mathbf{c}_n^1)} - 1 \right).$$

Finally, it follows that

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} \geq \lim_{n \rightarrow +\infty} \frac{n}{2} - \frac{1}{n-1} = +\infty.$$

□

4 Theoretical comparison between the first two upper bounds

In this section, we provide two sequences of EWMCP instances in which the ratio between the two upper bounds $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$, or its reciprocal, becomes arbitrarily large as the number of vertices in the graphs tends to infinity.

4.1 The second upper bound versus the first

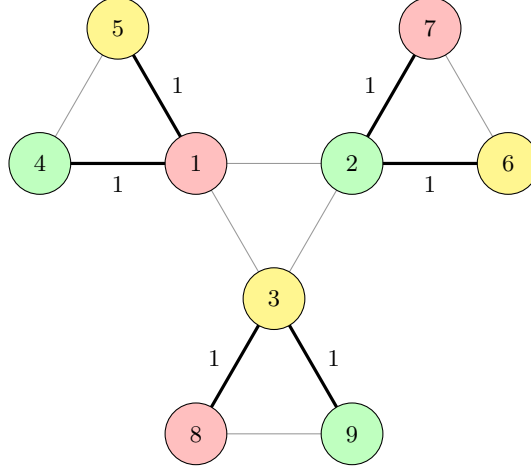
For every integer $n \geq 2$, let $G_n^2 = (V_n^2, E_n^2)$ be a graph with vertex set containing a set C_0 of n vertices, and, for each vertex $u \in C_0$, a set C_u of additional $n-1$ vertices, with $C_u \cap C_0 = \emptyset$. Moreover, the sets C_u for every $u \in C_0$ are pairwise disjoint. The edge set contains only the edges that make C_0 and $C_u \cup \{u\}$, for every $u \in C_0$, cliques. Hence, the graph is composed of n *peripheral cliques* $C_u \cup \{u\}$ of size n , each sharing exactly one vertex u with the *central clique* C_0 . For every edge $e \in E$, its edge-weight is defined as follows:

$$c_e = \begin{cases} 1 & \text{if } u \in C_0 \text{ and } v \in C_u, \\ 0 & \text{otherwise.} \end{cases}$$

That is, the edges connecting the central clique to the peripheral cliques have weight 1, and all the remaining edges (within each clique) have weight 0. These edge-weights define the edge-weight vector $\mathbf{c}_n^2$ for graph G_n^2 . An illustration of a graph of this family, with $n = 3$, is shown in Figure 4.

A coloring $\mathcal{C}_n^2$ of the vertices of G_n^2 with the minimum number n of colors can be constructed as follows. For all $u \in C_0$, an independent set I_u is created containing the vertex u and one vertex for every set C_v with $v \neq u$, which has not been included in an independent set yet. By construction, every edge of weight 1 joins vertices in different independent sets in $\mathcal{C}_n^2$. Additionally, each I_u is an independent set of cardinality n and $\mathcal{C}_n^2 := \{I_u : u \in C_0\}$ is indeed a coloring of G_n^2 .

Figure 4: Representation of the graph G_3^2 , where the vertices are colored using the minimum number of colors. Each vertex is labeled with a natural number displayed inside the node. Edges of weight 1 are shown in black with their weight displayed next to them; edges of weight 0 are shown in gray.



The following proposition states that the ratio between the upper bound $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ and $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ can be arbitrarily large.

Proposition 4.1. *The sequence $\{(G_n^2, \mathbf{c}_n^2)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^2$ is the coloring of G_n^2 defined above and $\prec_n^2$ defines any total order of the independent sets in $\mathcal{C}_n^2$.

Proof. We now define a feasible solution to Model (3) for the instance $(G_n^2, \mathbf{c}_n^2)$, leading to an upper bound on $\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)$. We assign the weight of every edge of weight 1 entirely on its endpoint in the peripheral clique, namely

$$\tilde{\varrho}_{eu} = 0, \quad \tilde{\varrho}_{ev} = 1, \quad u \in C_0, v \in C_u.$$

We have that each vertex in the peripheral cliques receives a total weight equal to 1, while each vertex in the central clique receives a total weight equal to 0. Hence, the feasible solution is completed by setting

$$\tilde{\pi}_I = 1, \quad I \in \mathcal{C}_n^2.$$

Therefore

$$\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2) \leq \sum_{I \in \mathcal{C}_n^2} \tilde{\pi}_I = n.$$

Let us now compute the value of $\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2)$. We know that, for each vertex in the

central clique $u \in \{1, 2, \dots, n\}$, belonging to the independent set $I_u \in \mathcal{C}_n^2$, there is exactly one neighbor of weight 1 in each of the independent sets I_v with $I_v \prec_n^2 I_u$. Hence, for every $u \in \{1, 2, \dots, n\}$, we have:

$$\sigma_u = |\{I_v \in \mathcal{C}_n^2 : I_v \prec_n^2 I_u\}|.$$

On the other hand, each vertex in the peripheral cliques $v \notin C_0$ has at most one incident edge of weight 1, hence $\sigma_v \leq 1$.

Therefore, for all independent sets I_u with $u \in \{1, 2, \dots, n\}$, the maximum $\max\{\sigma_v : v \in I_u\}$ is attained by the central vertex u . Then, for every $u \in \{1, 2, \dots, n\}$, we have:

$$\max\{\sigma_v : v \in I_u\} = |\{I_v \in \mathcal{C}_n^2 : I_v \prec_n^2 I_u\}|,$$

and thus, since $\prec_n^2$ is a total order

$$\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2) = \sum_{u=1}^n |\{I_v \in \mathcal{C}_n^2 : I_v \prec_n^2 I_u\}| = \sum_{h=0}^{n-1} h = \frac{n(n-1)}{2}.$$

Finally, it follows that

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} \geq \lim_{n \rightarrow +\infty} \frac{\frac{n(n-1)}{2}}{n} = \lim_{n \rightarrow +\infty} \frac{n-1}{2} = +\infty.$$

□

4.2 The first upper bound versus the second

For every integer $n \geq 1$, let $G_n^3 = (V_n^3, E_n^3)$ be a graph where the vertex set V_n^3 has cardinality $2n$ and it is partitioned into two independent sets $I = \{1, 2, \dots, n\}$ and $J = \{n+1, n+2, \dots, 2n\}$, each of cardinality n . By construction, no edges exist between vertices within the same independent set, while each vertex in I is connected to every vertex in J , i.e., for each $u \in I$, $v \in J$, there exists an edge $e \in E$ connecting u and v . To each edge $e \in E_n^3$ we assign the edge-weight $c_e = 1$. These edge-weights define the edge-weight vector $\mathbf{c}_n^3$ for the graph G_n^3 . An illustration of the graph G_n^3 is provided in Figure 5.

A coloring of the vertices of G_n^3 with the minimum number of colors is naturally given as $\mathcal{C}_n^3 = \{I_1, I_2\}$.

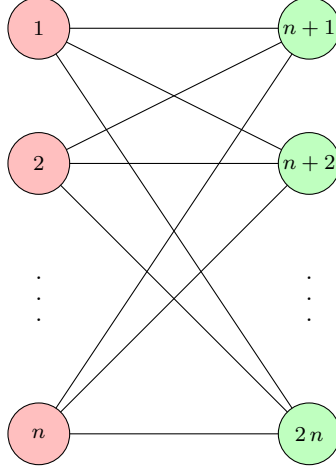
The following proposition states that the ratio between the upper bound $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ can be arbitrarily large.

Proposition 4.2. *The sequence $\{(G_n^3, \mathbf{c}_n^3)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)}{\text{UB}_2(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3, \prec_n^3)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^3$ is the coloring of G_n^3 defined above and $\prec_n^3$ defines any total order of the independent

Figure 5: Representation of the graph G_n^3 , where the vertices are colored using the minimum number of colors. Each vertex is labeled with a natural number displayed inside the node. All edge-weights are equal to 1; these values are omitted from the figure.



sets in $\mathcal{C}_n^3$.

Proof. We first show that the optimal value of Model (3), namely $\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)$, is n . We notice that, in any feasible solution of Model (3), for every $u \in I$ and $v \in J$, Constraints (3c) relative to the edge $e = \{u, v\}$ write as

$$\pi_I \geq \sum_{e \in \delta(u)} \varrho_{eu}, \quad \pi_J \geq \sum_{e \in \delta(v)} \varrho_{ev}.$$

Adding up these constraints over all vertices, therefore, yields

$$n(\pi_I + \pi_J) \geq \sum_{e \in E} c_e,$$

where we used Constraint (3b).

Moreover, all edge-weights are equal to 1, so the sum on the right-hand side equals $|E_n^3| = n^2$. Consequently, $\pi_I + \pi_J \geq n$, and every feasible solution of Model (3) has value at least n .

This lower bound of n on $\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)$ is tight: if we assign each edge-weight wholly to its endpoint in J and make the variable assignment $\pi_I = 0$, $\pi_J = n$, all constraints of Model (3) are satisfied and the objective function equals n .

Hence

$$\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3) = n.$$

As far as the bound $\text{UB}_2(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3, \prec_n^3)$ is concerned, we assume, without loss of generality that $I \prec_n^3 J$: observe that every vertex $u \in I$ is such that $\sigma_u = 0$, since there are no independent sets preceding I under the total order $\prec_n^3$. Whereas every vertex $v \in J$ is such that $\sigma_v = 1$,

since it is connected only to vertices in I with edges of weight 1. Hence, we have:

$$\max\{\sigma_u : u \in I\} = 0, \quad \max\{\sigma_v : v \in J\} = 1,$$

giving the upper bound

$$\text{UB}_2(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3, \prec_n^3) = 1.$$

Finally, it follows that

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)}{\text{UB}_2(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3, \prec_n^3)} = \lim_{n \rightarrow +\infty} \frac{n}{1} = +\infty.$$

□

Note that the values of the two upper bounds in the examples studied in this section, and thus the conclusion of the two propositions, depend on the choice of a particular coloring of the vertices of the graphs G_n^2 and G_n^3 . Nevertheless, our choices feature a minimum number of colors and the results hold with any coloring with this natural property.

5 A third upper bound on $\omega(G, \mathbf{c})$

In this section, we introduce and theoretically study a third upper bound on the edge-weighted clique number $\omega(G, \mathbf{c})$, originally proposed in Hosseini et al. (2020) and used within the branch-and-bound algorithm for the EWMCP detailed in the same article.

This bound requires an integer upper bound k on the cardinality of any clique in G , namely on the clique number $\omega(G)$ of the graph, with $1 \leq k \leq |V|$. Additionally, for all $u \in V$ and $t \in \mathbb{Z}_{\geq 0}$, we denote by $\delta_t(u) \subseteq \delta(u)$ any set of the $\min\{t, |\delta(u)|\}$ heaviest edges incident to u ; in particular, $\delta_0(u) = \emptyset$.

For all $u \in V$, we define the quantity

$$\eta_u := \frac{1}{2} \sum_{e \in \delta_{k-1}(u)} c_e, \tag{10}$$

which can be interpreted as an upper bound on the contribution of edges incident to vertex u to the total edge-weight of any clique of cardinality at most k containing u , since such a clique can contain at most $k - 1$ edges incident to u . The factor $\frac{1}{2}$ accounts for the fact that each edge of a clique is incident to two of its vertices. This value, similarly to γ_u , defined in (4), and σ_u , defined in (6), can be also interpreted as the total weight assigned to vertex u .

Finally, let $\varphi(j)$, with $j \in \{1, 2, \dots, |V|\}$, be a function that returns the vertex u with the j -th largest η_u , with ties broken arbitrarily. The function φ induces the following ordering on the vertex quantities (10):

$$\eta_{\varphi(1)} \geq \eta_{\varphi(2)} \geq \dots \geq \eta_{\varphi(|V|)}.$$

The third upper bound studied in this article, denoted by $\text{UB}_3(G, \mathbf{c}, k)$, is then defined as:

$$\text{UB}_3(G, \mathbf{c}, k) := \sum_{j=1}^k \eta_{\varphi(j)}, \quad (11)$$

that is the sum of the weights η_u of the k heaviest vertices of V .

For completeness, we verify that $\text{UB}_3(G, \mathbf{c}, k)$ is a valid upper bound on $\omega(G, \mathbf{c})$. Consider any clique C in G ; since k is an upper bound on the clique number of G , we have $|C| \leq k$. Then:

$$\sum_{e \in E[C]} c_e = \sum_{u \in C} \frac{1}{2} \sum_{v \in C \setminus \{u\}} c_{\{u,v\}} \quad (12)$$

$$\leq \sum_{u \in C} \frac{1}{2} \sum_{e \in \delta_{k-1}(u)} c_e \quad (13)$$

$$= \sum_{u \in C} \eta_u \quad (14)$$

$$\leq \sum_{j=1}^k \eta_{\varphi(j)} = \text{UB}_3(G, \mathbf{c}, k). \quad (15)$$

The equality in (12) rewrites the total edge-weight of the clique C by distributing each edge-weight equally between its two endpoints. The inequality in (13) follows from the fact that $|C \setminus \{u\}| \leq k - 1$. Therefore, the total weight of the edges connecting u to the other vertices of C is bounded above by the sum of the weights of the $k - 1$ heaviest edges incident to u . The equality in (14) holds by the definition (10) of η_u . Finally, the inequality in (15) follows from $|C| \leq k$, which implies that the sum of η_u over the vertices of C is bounded above by the sum of the k largest values of η_u over all vertices in V .

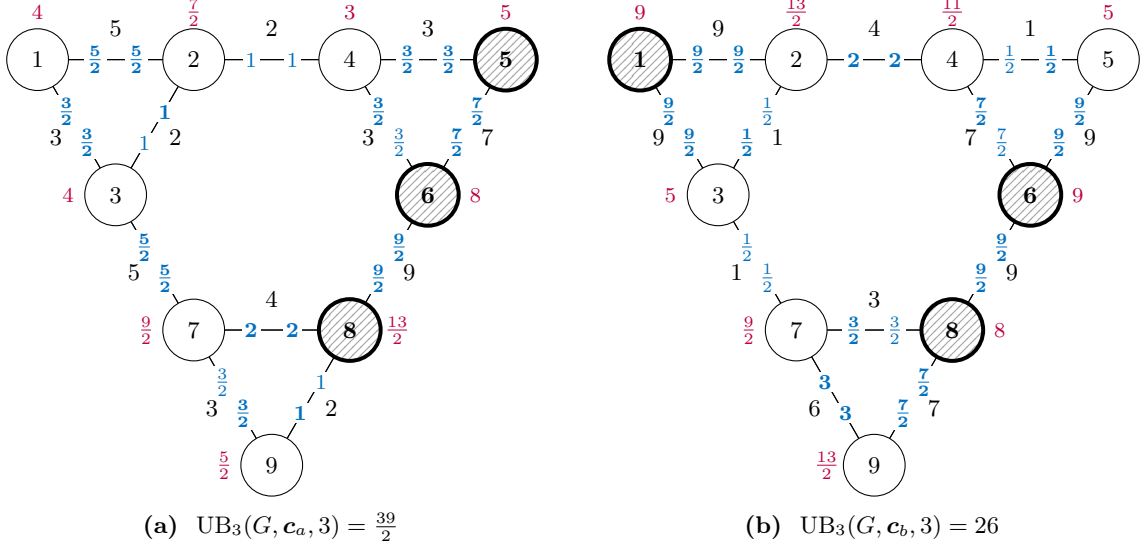
In Figure 6, we report the values of the upper bound $\text{UB}_3(G, \mathbf{c}, k)$ for the two instances introduced in Figure 1, with $k = 3$.

5.1 Theoretical comparison between the second and third upper bounds

In this subsection, we theoretically compare $\text{UB}_3(G, \mathbf{c}, k)$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$. While $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ depends on both the coloring $\mathcal{C}$ and the ordering of its independent sets, $\text{UB}_3(G, \mathbf{c}, k)$ depends only on k , an upper bound on the clique number of G . Since any coloring $\mathcal{C}$ provides an upper bound $|\mathcal{C}|$ on the clique number of G , we set $k = |\mathcal{C}|$ to compare the two upper bounds. We remark that a tighter upper bound on the clique number of G would yield a smaller k and thus a stronger $\text{UB}_3(G, \mathbf{c}, k)$. The following proposition states that, under the choice $k = |\mathcal{C}|$, $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ is always within a factor of 2 of $\text{UB}_3(G, \mathbf{c}, k)$, independently of the total order $\prec$.

Proposition 5.1. *For every EWMCP instance $(G, \mathbf{c})$, for every coloring $\mathcal{C}$ of the vertices of G*

Figure 6: The values of the upper bound $\text{UB}_3(G, \mathbf{c}, k)$ for the instances in Figure 1, with $k = 3$. For each edge $e \in E$, the black label next to the edge shows its weight c_e — in Part 6a, this corresponds to the edge-weight vector $\mathbf{c}_a$ of Part 1a of Figure 1; in Part 6b, to the edge-weight vector $\mathbf{c}_b$ of Part 1b of Figure 1. For each edge $e \in E$, the value $\frac{c_e}{2}$ is reported in blue on each side of the edge. A blue label is shown in bold if the corresponding edge is among the $k - 1 = 2$ heaviest edges incident to that vertex, meaning it contributes to the vertex weight η_u . For each vertex $u \in V$, the value η_u , computed as the sum of the bold blue labels incident to u , is shown in purple next to the vertex. The $k = 3$ vertices with the largest η_u are highlighted in bold with a striped pattern; their weights are summed to obtain $\text{UB}_3(G, \mathbf{c}, k)$.



and for every total order $\prec$ of the independent sets of $\mathcal{C}$, it holds:

$$\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec) \leq 2 \text{UB}_3(G, \mathbf{c}, |\mathcal{C}|). \quad (16)$$

Proof. For every vertex $u \in V$, by the definition (6), the total weight assigned to u in $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ is:

$$\sigma_u = \sum_{I \in \mathcal{C} : I \prec I(u)} \max \{ c_e : e \in \delta(u), v \in I \},$$

which is a sum of at most $|\{I \in \mathcal{C} : I \prec I(u)\}| \leq |\mathcal{C}| - 1$ terms, each being the maximum edge-weight among edges connecting u to vertices in a given independent set.

Since the independent sets $I \prec I(u)$ are pairwise disjoint, the terms in σ_u are weights of edges incident to u with distinct endpoints. Hence, σ_u is bounded above by the sum of the $|\mathcal{C}| - 1$ heaviest weights of edges incident to u . Therefore, using the definition of η_u given in (10), we obtain:

$$\sigma_u \leq \sum_{e \in \delta_{|\mathcal{C}|-1}(u)} c_e = 2\eta_u. \quad (17)$$

It follows that, for every independent set $I \in \mathcal{C}$,

$$\max \{ \sigma_u : u \in I \} \leq 2 \max \{ \eta_u : u \in I \}.$$

For each $I \in \mathcal{C}$, let $u_I^* \in \arg \max\{\eta_u : u \in I\}$. Hence, by the definition of φ ,

$$\sum_{I \in \mathcal{C}} \eta_{u_I^*} \leq \sum_{j=1}^{|\mathcal{C}|} \eta_{\varphi(j)} = \text{UB}_3(G, \mathbf{c}, |\mathcal{C}|).$$

Combining the above inequalities, we obtain

$$\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec) = \sum_{I \in \mathcal{C}} \max\{\sigma_u : u \in I\} \leq 2 \sum_{I \in \mathcal{C}} \eta_{u_I^*} \leq 2 \text{UB}_3(G, \mathbf{c}, |\mathcal{C}|),$$

which concludes the proof. $\square$

To show that the factor of 2 in Proposition 5.1 is asymptotically tight, we construct a family of instances for which the ratio between the upper bound $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ and $\text{UB}_3(G, \mathbf{c}, k)$ converges to 2 as the number of vertices tends to infinity.

For every integer $k \geq 2$, let $G_k^4 = (V_k^4, E_k^4)$ be a graph where the vertex set V_k^4 is partitioned into k independent sets, each of cardinality $k - 1$, defining a coloring $\mathcal{C}_k^4$, and let $\prec_k^4$ be a total order on $\mathcal{C}_k^4$. Thus, the vertex set has cardinality $|V_k^4| = k(k - 1)$.

The edge set E_k^4 is defined as follows. For every pair of independent sets $I, J \in \mathcal{C}_k^4$ with $I \prec_k^4 J$, we select a different pair of vertices $\{v_{I,J}, v_{J,I}\}$, with vertex $v_{I,J} \in I$ and vertex $v_{J,I} \in J$, and connect them by an edge of weight 1. Hence, the $\binom{k}{2}$ edges of weight 1 form a perfect matching on V_k^4 , meaning that each vertex is incident to exactly one edge of positive weight.

Then, every pair of non-adjacent vertices belonging to distinct independent sets is connected by an edge of weight 0. These edge-weights define the edge-weight vector $\mathbf{c}_k^4$ for the graph G_k^4 .

Since each vertex belongs to an independent set of cardinality $k - 1$ and is adjacent to all vertices in the remaining $k - 1$ independent sets, it follows that $|\delta(v)| = (k - 1)^2$ for every $v \in V_k^4$, with exactly one edge in $\delta(v)$ with weight 1 and the remaining $(k - 1)^2 - 1$ having weight 0.

By construction, the coloring $\mathcal{C}_k^4$ uses the minimum number of independent sets for the graph G_k^4 . An illustration of a graph of the family G_k^4 for $k = 3$ is provided in Figure 7.

The following proposition shows that the ratio between the upper bound $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ and $\text{UB}_3(G, \mathbf{c}, k)$ can be arbitrarily close to 2, thus proving the asymptotic tightness of Proposition 5.1.

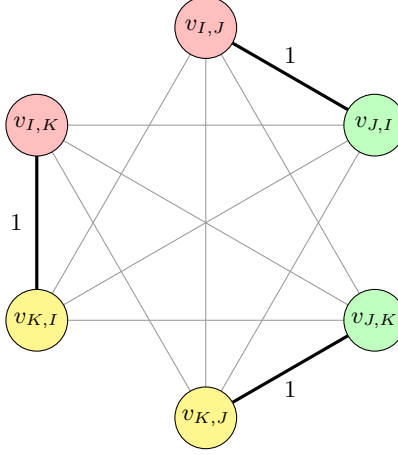
Proposition 5.2. *The sequence $\{(G_k^4, \mathbf{c}_k^4)\}_{k \geq 2}$ of EWMCP instances is such that*

$$\frac{\text{UB}_2(G_k^4, \mathbf{c}_k^4, \mathcal{C}_k^4, \prec_k^4)}{\text{UB}_3(G_k^4, \mathbf{c}_k^4, k)} \rightarrow 2 \quad \text{as} \quad k \rightarrow +\infty,$$

where $\mathcal{C}_k^4$ is the coloring of G_k^4 with cardinality k defined above and $\prec_k^4$ defines any total order of the independent sets in $\mathcal{C}_k^4$.

Proof. Let us first determine the value of $\text{UB}_3(G_k^4, \mathbf{c}_k^4, k)$. By construction of G_k^4 , each vertex

Figure 7: Representation of the graph G_3^4 , where the vertices are colored using the minimum number of colors $k = 3$. The label of each vertex is displayed inside the node. The three independent sets are $I = \{1, 6\}$ (red), $J = \{2, 3\}$ (green), and $K = \{4, 5\}$ (yellow). The total order is given by $I \prec J \prec K$. Edges of weight 1 (black) form a perfect matching on V_3^4 , with their weight shown next to them; edges of weight 0 (gray) ensure the existence of a clique of cardinality 3.



$v_{I,J} \in I$ is incident to exactly one edge of positive weight, namely the edge $\{v_{I,J}, v_{J,I}\}$ of weight 1, while all the other $(k-1)^2 - 1$ incident edges have weight 0. Therefore, by definition (10), for every $v_{I,J} \in V_k^4$ we have

$$\eta_{v_{I,J}} = \frac{1}{2} \sum_{e \in \delta_{k-1}(v_{I,J})} c_e = \frac{1}{2}.$$

Since $\eta_u = 1/2$ for every $u \in V_k^4$, it follows from definition (11) that

$$\text{UB}_3(G_k^4, \mathbf{c}_k^4, k) = \sum_{j=1}^k \eta_{\varphi(j)} = k \cdot \frac{1}{2} = \frac{k}{2}.$$

Let us now compute $\text{UB}_2(G_k^4, \mathbf{c}_k^4, \mathcal{C}_k^4, \prec_k^4)$. According to definition (7), this amounts to determining, for every independent set $I \in \mathcal{C}_k^4$, the quantity

$$\max\{\sigma_u : u \in I\},$$

that is the maximum total weight assigned to a vertex in I . We distinguish two cases:

- (i) Let I be the lower independent set in the order $\prec_k^4$. Then, there is no $J \in \mathcal{C}_k^4$ such that $J \prec_k^4 I$. Hence, by definition (6), for every vertex $v_{I,J} \in I$, the sum defining $\sigma_{v_{I,J}}$ is empty, and therefore

$$\sigma_{v_{I,J}} = 0.$$

Thus, $\max\{\sigma_u : u \in I\} = 0$.

- (ii) Let I be any other independent set under $\prec_k^4$. For every $J \in \mathcal{C}_k^4$ with $J \prec_k^4 I$, by construction there is a vertex $v_{I,J} \in I$ incident to the edge $\{v_{I,J}, v_{J,I}\}$ of weight 1, where $v_{J,I} \in J$. All other edges incident to $v_{I,J}$ have weight 0. Therefore, for every $v_{I,J} \in I$ with

$$J \prec_k^4 I,$$

$$\sigma_{v_{I,J}} = 1.$$

On the other hand, for every $K \in \mathcal{C}_k^4$ with $I \prec_k^4 K$, the unique positive-weight edge incident to $v_{I,K} \in I$ connects it to the vertex $v_{K,I}$ in K and therefore does not contribute to the sum in (6); hence

$$\sigma_{v_{I,K}} = 0.$$

It follows that $\max\{\sigma_u : u \in I\} = 1$.

Since $\mathcal{C}_k^4$ consists of k independent sets and exactly one of them is the first under the total order $\prec_k^4$, we obtain

$$\text{UB}_2(G_k^4, \mathbf{c}_k^4, \mathcal{C}_k^4, \prec_k^4) = \sum_{I \in \mathcal{C}_k^4} \max\{\sigma_u : u \in I\} = 0 + \underbrace{1 + 1 + \dots + 1}_{k-1 \text{ terms}} = k - 1.$$

Finally, the ratio between the two upper bounds is

$$\frac{\text{UB}_2(G_k^4, \mathbf{c}_k^4, \mathcal{C}_k^4, \prec_k^4)}{\text{UB}_3(G_k^4, \mathbf{c}_k^4, k)} = \frac{k-1}{\frac{k}{2}} = \frac{2(k-1)}{k} = 2 - \frac{2}{k},$$

and therefore

$$\lim_{k \rightarrow +\infty} \frac{\text{UB}_2(G_k^4, \mathbf{c}_k^4, \mathcal{C}_k^4, \prec_k^4)}{\text{UB}_3(G_k^4, \mathbf{c}_k^4, k)} = 2,$$

which concludes the proof. $\square$

We further provide a sequence of EWMCP instances in which the ratio between the upper bound $\text{UB}_3(G, \mathbf{c}, k)$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ becomes arbitrarily large as the number of vertices in the graphs tends to infinity.

For every integer $n \geq 2$, let $G_n^5 = (V_n^5, E_n^5)$ be a graph where the vertex set V_n^5 has cardinality $2n$ and is partitioned into an independent set $I \subseteq V_n^5$ and a clique $C \subseteq V_n^5$, where

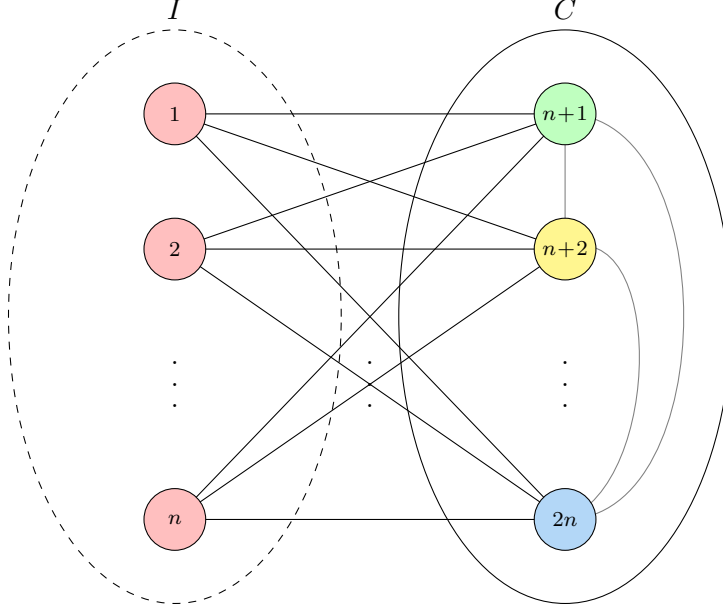
$$I = \{1, 2, \dots, n\}, \quad C = \{n+1, n+2, \dots, 2n\}.$$

The edge set E_n^5 of the graph includes all edges within the clique, each assigned an edge-weight of 0, and all edges connecting each vertex in I to every vertex in C , each assigned an edge-weight of 1. These edge-weights define the edge-weight vector $\mathbf{c}_n^5$ for the graph G_n^5 . An illustration of this graph is provided in Figure 8.

Since G_n^5 contains a clique of cardinality n , any coloring of its vertices requires at least n colors. Moreover, every vertex in I is adjacent to every vertex in C , so the vertices of I require at least one additional color, giving a total of at least $n+1$ colors. A coloring $\mathcal{C}_n^5$ of the vertices of G_n^5 with the minimum number $n+1$ of colors is given by

$$\mathcal{C}_n^5 = \{I, \{n+1\}, \{n+2\}, \dots, \{2n\}\}.$$

Figure 8: Representation of the graph G_n^5 , where the vertices are colored using the minimum number of colors. Each vertex is labeled with a natural number displayed inside the node. The dashed ellipse on the left represents the independent set I , while the ellipse on the right represents the clique C . Edges of weight 1 connecting a vertex in I to a vertex in C are shown in black, while edges of weight 0 within C are shown in gray. Edge-weight labels are omitted from the figure.



The following proposition states that the ratio between the upper bound $\text{UB}_3(G, \mathbf{c}, k)$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ can be arbitrarily large.

Proposition 5.3. *The sequence $\{(G_n^5, \mathbf{c}_n^5)\}_{n \geq 2}$ of EWMCP instances is such that*

$$\frac{\text{UB}_3(G_n^5, \mathbf{c}_n^5, n+1)}{\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^5$ is the coloring of G_n^5 with cardinality $n+1$ defined above and $\prec_n^5$ defines any total order of the independent sets in $\mathcal{C}_n^5$.

Proof. We first compute $\text{UB}_3(G_n^5, \mathbf{c}_n^5, n+1)$, with $k = |\mathcal{C}_n^5| = n+1$. Since $k-1 = n$, for every vertex $u \in V_n^5$, the set $\delta_n(u)$ consists of the n heaviest edges incident to u .

For every vertex $u \in I$, the vertex u has exactly n incident edges (one to each vertex of C), all of weight 1. Therefore, for $u \in I$, by definition (10),

$$\eta_u = \frac{1}{2} \sum_{e \in \delta_n(u)} c_e = \frac{1}{2} \cdot n \cdot 1 = \frac{n}{2}.$$

For every vertex $u \in C$, the vertex u has n incident edges to the vertices of I , each of weight 1, and $n-1$ incident edges to the other vertices of C , each of weight 0. Thus, $\delta_n(u)$ consists of the

n edges connecting u to each vertex of I , and

$$\eta_u = \frac{1}{2} \sum_{e \in \delta_n(u)} c_e = \frac{1}{2} \cdot n \cdot 1 = \frac{n}{2}.$$

Since $\eta_u = n/2$ for every $u \in V_n^5$, it follows from definition (11) that

$$\text{UB}_3(G_n^5, \mathbf{c}_n^5, n+1) = \sum_{j=1}^{n+1} \eta_{\varphi(j)} = (n+1) \cdot \frac{n}{2} = \frac{n(n+1)}{2}.$$

We now compute $\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5)$ for an arbitrary total order $\prec_n^5$.

We first determine σ_u for every vertex $u \in I$. Suppose that the independent set I occupies position p , with $p \in \{1, 2, \dots, n+1\}$, in the total order $\prec_n^5$, so that I is preceded by exactly $p-1$ independent sets, each containing a single vertex of C .

Since every vertex $u \in I$ is connected to every vertex of C by an edge of weight 1, each independent set $J \prec_n^5 I$ contributes 1 to the sum in definition (6). Hence, for every $u \in I$,

$$\sigma_u = \sum_{J \in \mathcal{C}_n^5: J \prec_n^5 I} \max\{c_e : e \in \delta(u), v \in J\} = (p-1) \cdot 1 = p-1.$$

Let us now determine σ_u for every vertex $u \in C$. By definition of the coloring $\mathcal{C}_n^5$, each vertex $u \in C$ belongs to a distinct independent set $\{u\} \in \mathcal{C}_n^5$. We distinguish two cases, depending on whether $\{u\}$ precedes or follows I under the total order $\prec_n^5$.

If $\{u\}$ precedes I under the total order $\prec_n^5$, then only independent sets containing a single vertex of C precede $\{u\}$, each contributing 0 to the sum in definition (6), since all edges within C have weight 0. Hence, for every $u \in C$ with $\{u\} \prec_n^5 I$,

$$\sigma_u = \sum_{J \in \mathcal{C}_n^5: J \prec_n^5 \{u\}} \max\{c_e : e \in \delta(u), v \in J\} = 0.$$

If instead $I \prec_n^5 \{u\}$, then the independent set I contributes 1 to the sum in definition (6), since u is connected to every vertex of I by an edge of weight 1. All other preceding independent sets contain a single vertex of C , each contributing 0 as above. Hence, for every $u \in C$ with $I \prec_n^5 \{u\}$,

$$\sigma_u = \sum_{J \in \mathcal{C}_n^5: J \prec_n^5 \{u\}} \max\{c_e : e \in \delta(u), v \in J\} = 1.$$

Since exactly $p-1$ independent sets precede I and $n-p+1$ follow it in the total order $\prec_n^5$, there are $n-p+1$ vertices $u \in C$ with $I \prec_n^5 \{u\}$ and $p-1$ vertices $u \in C$ with $\{u\} \prec_n^5 I$. By

definition (7), we obtain

$$\begin{aligned}
\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5) &= \sum_{J \in \mathcal{C}_n^5} \max\{\sigma_u : u \in J\} \\
&= \max\{\sigma_u : u \in I\} + \sum_{\{u\} \in \mathcal{C}_n^5 \setminus \{I\}} \sigma_u \\
&= (p-1) + (n-p+1) \cdot 1 + (p-1) \cdot 0 \\
&= n.
\end{aligned}$$

Hence $\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5) = n$ for every total order $\prec_n^5$.

Finally, the ratio between the two upper bounds is

$$\frac{\text{UB}_3(G_n^5, \mathbf{c}_n^5, n+1)}{\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5)} = \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2},$$

and therefore

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_3(G_n^5, \mathbf{c}_n^5, n+1)}{\text{UB}_2(G_n^5, \mathbf{c}_n^5, \mathcal{C}_n^5, \prec_n^5)} = +\infty,$$

which concludes the proof. $\square$

5.2 Theoretical comparison between $\text{UB}_3(G, \mathbf{c}, k)$ and $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$

We show that either of the two upper bounds $\text{UB}_3(G, \mathbf{c}, k)$ and $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ can be arbitrarily worse than the other, by using the families of instances $\{(G_n^2, \mathbf{c}_n^2)\}$ and $\{(G_n^3, \mathbf{c}_n^3)\}$ introduced in Section 4.

Proposition 5.4. *The sequence $\{(G_n^2, \mathbf{c}_n^2)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_3(G_n^2, \mathbf{c}_n^2, n)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^2$ is the coloring of G_n^2 with cardinality n , defined in Section 4.

Proof. By Theorem 4.1, for any total order $\prec_n^2$ of the independent sets of $\mathcal{C}_n^2$, we have

$$\frac{\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty.$$

By Theorem 5.1, with $k = |\mathcal{C}_n^2| = n$ it holds that, for every $n \geq 1$,

$$\text{UB}_3(G_n^2, \mathbf{c}_n^2, n) \geq \frac{1}{2} \text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2).$$

Therefore,

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_3(G_n^2, \mathbf{c}_n^2, n)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} \geq \lim_{n \rightarrow +\infty} \frac{1}{2} \frac{\text{UB}_2(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2, \prec_n^2)}{\text{UB}_1(G_n^2, \mathbf{c}_n^2, \mathcal{C}_n^2)} = +\infty.$$

□

Proposition 5.5. *The sequence $\{(G_n^3, \mathbf{c}_n^3)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)}{\text{UB}_3(G_n^3, \mathbf{c}_n^3, 2)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where $\mathcal{C}_n^3$ is the coloring of G_n^3 with cardinality 2, defined in Section 4.

Proof. As shown in the proof of Theorem 4.2, $\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3) = n$.

Let us now compute $\text{UB}_3(G_n^3, \mathbf{c}_n^3, 2)$. Let $k = 2 = |\mathcal{C}_n^3|$. Since $k = 2$, for every vertex $u \in V_n^3$, the set $\delta_{k-1}(u) = \delta_1(u)$ contains a single edge, namely the heaviest edge incident to u . Since all edge-weights are equal to 1, by definition (10), we have

$$\eta_u = \frac{1}{2} \sum_{e \in \delta_1(u)} c_e = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

for every $u \in V_n^3$. It follows from definition (11) that

$$\text{UB}_3(G_n^3, \mathbf{c}_n^3, 2) = \sum_{j=1}^k \eta_{\varphi(j)} = 2 \cdot \frac{1}{2} = 1.$$

Therefore,

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_1(G_n^3, \mathbf{c}_n^3, \mathcal{C}_n^3)}{\text{UB}_3(G_n^3, \mathbf{c}_n^3, 2)} = \lim_{n \rightarrow +\infty} \frac{n}{1} = +\infty.$$

□

5.3 Theoretical comparison between $\text{UB}_3(G, \mathbf{c}, k)$ and $\omega(G, \mathbf{c})$

Finally, the following proposition shows that the ratio between $\text{UB}_3(G, \mathbf{c}, k)$ and the edge-weighted clique number $\omega(G, \mathbf{c})$ can be arbitrarily large, by considering the family of instances $\{(G_n^1, \mathbf{c}_n^1)\}$ introduced in Section 3.

Proposition 5.6. *The sequence $\{(G_n^1, \mathbf{c}_n^1)\}$ of EWMCP instances is such that*

$$\frac{\text{UB}_3(G_n^1, \mathbf{c}_n^1, n)}{\omega(G_n^1, \mathbf{c}_n^1)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty,$$

where n coincides with the clique number of G_n^1 .

Proof. By Proposition 3.2, for any coloring $\mathcal{C}_n^1$ of G_n^1 and any total order $\prec_n^1$ of the independent sets of $\mathcal{C}_n^1$, we have

$$\frac{\text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} \rightarrow +\infty \quad \text{as} \quad n \rightarrow +\infty.$$

By construction, the minimum possible number of colors for G_n^1 is n ; let $\mathcal{C}_n^1$ be a coloring with

$|\mathcal{C}_n^1| = n$. By Proposition 5.1 with $k = |\mathcal{C}_n^1| = n$, it holds that, for every $n \geq 1$,

$$\text{UB}_3(G_n^1, \mathbf{c}_n^1, n) \geq \frac{1}{2} \text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1).$$

Therefore,

$$\lim_{n \rightarrow +\infty} \frac{\text{UB}_3(G_n^1, \mathbf{c}_n^1, n)}{\omega(G_n^1, \mathbf{c}_n^1)} \geq \lim_{n \rightarrow +\infty} \frac{1}{2} \frac{\text{UB}_2(G_n^1, \mathbf{c}_n^1, \mathcal{C}_n^1, \prec_n^1)}{\omega(G_n^1, \mathbf{c}_n^1)} = +\infty.$$

□

6 Computational comparison

In this section, we report the results of our extensive experimental analysis, conducted on both DIMACS instances and randomly generated instances commonly used in the literature for the Edge-Weighted Maximum Clique Problem (EWMCP). This computational campaign provides practical insights into the empirical strength of the upper bounds on $\omega(G, \mathbf{c})$ analyzed in this work: $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$, proposed by San Segundo et al. (2019) and described in Section 2.1, $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$, proposed by Shimizu et al. (2020) and described in Section 2.2, and $\text{UB}_3(G, \mathbf{c}, k)$, proposed by Hosseinian et al. (2020) and described in Section 5.

The purpose of our tests is twofold. On the one hand, we aim to determine which of the upper bounds is closer to $\omega(G, \mathbf{c})$ in practice. On the other hand, we seek to assess whether their empirical strength depends on instance characteristics (such as the number of vertices and graph density) or on algorithmic choices (such as the type of coloring and the ordering of its independent sets).

Both $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$ depend on a coloring $\mathcal{C}$ of the instance graph G . In our computational experiments, such coloring is obtained using the DSatur heuristic (Br  laz, 1979). In the following, we refer to a coloring obtained via the DSatur algorithm as *DSatur coloring*. Unlike the other two upper bounds, $\text{UB}_3(G, \mathbf{c}, k)$ depends on an upper bound k on the clique number of G , instead of a coloring of the graph. In order to ensure a fair comparison among the three bounds studied in this paper, we always set k to be equal to the cardinality of the same coloring $\mathcal{C}$ used by $\text{UB}_1(G, \mathbf{c}, \mathcal{C})$ and $\text{UB}_2(G, \mathbf{c}, \mathcal{C}, \prec)$. For the sake of brevity, in the remainder of this section we refer to the upper bounds simply as UB_1 , UB_2 and UB_3 .

All experiments are conducted on a Linux machine equipped with a 2.30 GHz CPU and 512 GB of RAM. The code is implemented in C++, and we use IBM ILOG CPLEX 22.11 to compute UB_1 , as it requires solving the LP Model (3). Extensive preliminary tests show that the most efficient algorithm for this purpose is the barrier method, without performing the crossover phase, since we are only interested in the value of the optimal solution of Model (3). Indeed, our computational experiments demonstrate that the barrier algorithm solves the problem to optimality in a time that is orders of magnitude lower than that required by alternative LP solvers, such as the primal simplex, dual simplex, or network simplex methods.

Detailed numerical results, beyond those presented in the paper, along with the source code for computing the upper bounds, are available at the following GitHub repository: <https://github.com>.

[com/FabioCiccarelli/EWMCP_Bounds.git](https://github.com/FabioCiccarelli/EWMCP_Bounds). This material supports precise benchmarking and facilitates comparisons with alternative approaches. We also hope that our findings and the accompanying resources will stimulate further research on upper bounds for $\omega(G, \mathbf{c})$, with the aim of identifying tighter values that can enhance the performance of bounding procedures in exact branch-and-bound algorithms for the EWMCP.

The remainder of this section is organized as follows. In Section 6.1, we describe the benchmark datasets used in the experiments. In Section 6.2, we analyze the impact of the ordering of the independent sets on the quality of UB_2 . In Section 6.3 and Section 6.4, we report the results on DIMACS and RANDOM instances, respectively. In Section 6.5, we assess the behavior of the three bounds in a simulated branch-and-bound setting. Finally, in Section 6.6, we compare UB_1 with other LP-based upper bounds from the literature.

6.1 Benchmark datasets of instances

In this work, we consider both DIMACS instances and randomly generated (RANDOM) instances.

- The DIMACS instances, see DIMACS (2017), consist of 80 graphs that are commonly used in the literature to evaluate the performance of exact algorithms for the EWMCP, such as those proposed by Shimizu et al. (2020), San Segundo et al. (2019) and Hosseinian et al. (2020). These algorithms are able to determine $\omega(G, \mathbf{c})$ for 48 of the 80 instances, making this benchmark a valuable reference for assessing the strength of upper bounds.
- The RANDOM instances are generated according to specific pairs of number of vertices and edge density $(|V|, \mu)$, following common practice in the literature. For each combination of $|V| \in \{10, 20, \dots, 100\}$ and $\mu \in \{0.1, 0.2, \dots, 0.9\}$, we generate 10 instances. Additionally, for the case $|V| = 100$, we include higher densities in the range $[0.91, 0.99]$ with a step size of 0.01, resulting in a total of 990 graphs. Notice that the 10 instances for each combination $(|V|, \mu)$ are generated independently from each other. These instances are designed to evaluate the influence of instance characteristics on upper bound quality. Graphs with up to 100 vertices are considered, as they can be solved to optimality by state-of-the-art exact methods, allowing for a direct comparison with $\omega(G, \mathbf{c})$. Smaller graphs are included to assess the impact of the number of vertices, while high-density graphs (91–99%) are included due to their well-known difficulty for exact EWMCP algorithms.

Edge-weights are assigned following a standard approach commonly adopted in the literature Gouveia and Martins (2015); Hosseinian et al. (2018); Shimizu et al. (2019); San Segundo et al. (2019). Specifically, for each edge $e \in E$, the edge-weight is defined as $c_e := ((u+v) \bmod 200) + 1$.

6.2 Sensitivity of $UB_2(G, \mathbf{c}, \mathcal{C}, \prec)$ to the ordering of the independent sets

As discussed in Section 2.2, $UB_2(G, \mathbf{c}, \mathcal{C}, \prec)$ depends on the total order $\prec$ of the independent sets of $\mathcal{C}$: different orderings yield different vertex weights, as defined in (6) and, consequently, different bound values.

We compare five ordering criteria to determine whether any of them consistently minimizes

the value of UB_2 : *natural ordering* (NAT), in which the independent sets are kept in the order returned by the coloring algorithm (e.g., DSatur); *non-decreasing size* (SZ $\uparrow$) and *non-increasing size* (SZ $\downarrow$), in which independent sets are sorted by their cardinality; *non-decreasing star-weight score* (SW $\uparrow$) and *non-increasing star-weight score* (SW $\downarrow$), in which independent sets are sorted by the average, over their vertices, of the total weight of the $(k-1)$ heaviest edges incident to each vertex $u \in V$, i.e., the total weight of the edges in $\delta_{k-1}(u)$ as defined in Section 5.

The size-based orderings are motivated by the observation that the cardinality of each independent set determines the number of candidate vertices competing for the maximum in (6), so placing larger or smaller sets earlier may affect the bound. The star-weight score provides instead a proxy for the potential contribution of each independent set to the overall bound.

Table 1 compares the five strategies on the DIMACS instances, reporting, for each strategy and instance family, the number of instances for which the strategy yields the tightest upper bound (#best) and the average percentage increase of the bound relative to the minimum over all five orderings (% diff.). All the instances have been tested using DSatur as a coloring algorithm. Sorting the independent sets by non-decreasing cardinality (SZ $\uparrow$) dominates the other strategies: it achieves the lowest overall average deviation from the minimum (0.12%) and yields the tightest bound on 72 of the 80 instances. The ordering by non-decreasing star weight score (SW $\uparrow$) is the second-best alternative, with an average deviation of 3.68% from the minimum bound, though it is preferable to SZ $\uparrow$ only for the MANN family. The remaining orders (NAT, SZ $\downarrow$, SW $\downarrow$) yield weaker bounds, with average deviations ranging from 4.49% to 6.83%. Based on these results, throughout the remainder of the computational section we compute UB_2 by sorting the independent sets of the coloring in non-decreasing order of cardinality.

Table 1: Average percentage increase over the best ordering (% diff.) and number of best results (#best) per sorting strategy, on DIMACS instances, grouped by instance family. The upper bound is computed using DSatur coloring.

Instances		NAT		SZ $\uparrow$		SZ $\downarrow$		SW $\uparrow$		SW $\downarrow$	
family	#	% diff.	#best	% diff.	#best	% diff.	#best	% diff.	#best	% diff.	#best
brock	12	8.63	0	0.00	12	9.06	0	3.02	0	6.10	0
C	7	5.41	0	0.00	7	6.31	0	3.49	0	5.05	0
c-fat	7	6.23	0	0.65	5	7.24	0	5.85	1	5.82	1
DSJC	2	8.59	0	0.00	2	9.06	0	4.40	0	5.81	0
gen	5	8.21	0	0.00	5	10.38	0	5.08	0	6.00	0
hamming	6	3.60	1	0.00	6	4.70	1	3.94	1	3.26	1
johnson	4	11.25	0	0.00	4	11.35	0	8.88	0	6.68	0
keller	3	8.73	0	0.00	3	11.86	0	6.66	0	6.11	0
MANN	4	2.42	0	1.04	1	0.92	3	1.09	0	1.79	0
p_hat	15	4.25	0	0.00	15	5.17	0	3.20	0	2.74	0
san	11	2.72	1	0.09	8	3.68	0	1.68	0	2.04	2
sanr	4	10.11	0	0.00	4	10.52	0	2.18	0	8.30	0
Tot/Avg	80	6.00	2	0.12	72	6.83	4	3.68	2	4.49	4

6.3 Results for DIMACS instances

In this section, we present the results of the tests conducted on the DIMACS instances. Throughout this section, all bounds are computed using a DSatur coloring $\mathcal{C}$. Specifically, UB_1 and UB_2 use $\mathcal{C}$ directly, with the independent sets of $\mathcal{C}$ sorted in non-decreasing order of cardinality for UB_2 ; UB_3 uses instead $k = |\mathcal{C}|$.

Table 2 reports detailed results for each individual DIMACS instance. For each instance, the table lists its name, number of vertices $|V|$, number of edges $|E|$, the clique number $\omega(G)$ of the instance graph (when known), and the value of $\omega(G, c)$ (when known). The cardinality of the DSatur coloring of the instance graph, as well as the average size of its independent sets, are reported in the following two columns. Finally, the last columns show the value of each of the three upper bounds. For each instance, the smaller of the upper bound values is highlighted in bold. The results reported in the table clearly show that UB_2 outperforms UB_1 and UB_3 on nearly all DIMACS instances, with the only exceptions being **c-fat200-1** and **c-fat200-5**, where UB_1 achieves the smallest value among the three bounds. Furthermore, we observe a correlation between the number of vertices $|V|$ and the relative tightness of the bounds. Specifically, for instances with a similar edge density, the relative gap between UB_1 and UB_2 widens as the graph size increases. This behavior is evident within the **C** instance family (with a constant density of roughly 0.9), where the ratio UB_1/UB_2 grows from approximately 1.45 for **C125.9** to nearly 1.90 for **C2000.9**, indicating that UB_2 scales better on larger topologies. Graph density also plays a crucial role in the relative performance of the bounds. Observing the **san** instances, it can be noted that the advantage of UB_2 over UB_1 is remarkably pronounced on medium-density graphs. For example, on **san400_0.5_1** (density 0.5), the value of UB_1 is more than seven times larger than that of UB_2 . However, as density increases towards 0.9 (e.g., **san400_0.9_1**), this divergence narrows considerably, although UB_2 still maintains a clear superiority. The results shown in Table 2 also highlight a correlation between the performance of UB_2 and that of UB_3 . In the vast majority of the test set, UB_3 is slightly higher than UB_2 , consistently yielding tighter bound values than UB_1 . Noticeable deviations between UB_2 and UB_3 emerge primarily on exceptionally dense or massive graphs, such as those belonging to the **MANN** family, where the differences between the two bounds become more apparent. Nevertheless, despite the dominance of UB_2 over UB_1 and UB_3 , it is important to note that all the upper bounds are generally weak and often far from $\omega(G, c)$. On average, the ratio between bound value and the optimal value exceeds the value of 10 for all the considered upper bounds, highlighting their limited tightness. Nevertheless, UB_2 achieves an average ratio slightly above 11, less than half than that of UB_1 , and around 15% smaller than that of UB_3 .

Table 2: Values of the three upper bounds on DIMACS instances. A DSatur coloring $\mathcal{C}$ is used for computing UB_1 and UB_2 . For UB_2 , the independent sets of $\mathcal{C}$ are sorted in non-decreasing order of cardinality. UB_3 uses $k = |\mathcal{C}|$.

Instances					DSatur coloring		UB_1	UB_2	UB_3
name	$ V $	$ E $	$\omega(G)$	$\omega(G, c)$	card	avg size			
brock200_1	200	14,834	21	21,230	55	3.64	346,395	206,229	248,468
brock200_2	200	9,876	12	6,542	34	5.88	139,041	80,625	96,843

brock200_3	200	12,048	15	10,303	41	4.88	209,408	121,862	141,200
brock200_4	200	13,089	17	13,967	50	4.00	271,661	163,015	204,973
brock400_1	400	59,723	27	—	98	4.08	1,277,900	679,356	808,989
brock400_2	400	59,786	29	—	99	4.04	1,276,930	689,958	824,164
brock400_3	400	59,681	31	46,785	98	4.08	1,269,990	692,312	809,756
brock400_4	400	59,765	33	54,304	100	4.00	1,286,460	717,069	839,816
brock800_1	800	207,505	23	—	143	5.59	3,218,480	1,524,480	1,779,610
brock800_2	800	208,166	24	—	146	5.48	3,351,520	1,591,550	1,849,540
brock800_3	800	207,333	25	—	142	5.63	3,183,140	1,498,410	1,755,070
brock800_4	800	207,643	26	—	142	5.63	3,264,490	1,513,150	1,758,500
C125.9	125	6,963	34	66,248	52	2.40	279,002	192,784	227,809
C250.9	250	27,984	44	—	98	2.55	928,441	637,829	778,450
C500.9	500	112,332	—	—	175	2.86	3,458,980	2,159,570	2,547,210
C1000.9	1,000	450,079	—	—	316	3.16	12,581,900	7,058,440	8,277,240
C2000.5	2,000	999,836	16	—	223	8.97	9,876,680	3,894,920	4,461,250
C2000.9	2,000	1,799,532	—	—	586	3.41	47,423,100	24,977,100	28,891,500
C4000.5	4,000	4,000,268	18	—	399	10.03	35,853,700	12,836,300	14,447,900
c-fat200-1	200	1,534	12	7,734	15	13.33	10,677	11,226	14,115
c-fat200-2	200	3,235	24	26,389	24	8.33	36,326	31,203	37,990
c-fat200-5	200	8,473	58	168,200	84	2.38	304,982	347,871	363,583
c-fat500-1	500	4,459	14	10,738	14	35.71	14,422	11,982	14,441
c-fat500-2	500	9,139	26	38,350	26	19.23	50,942	40,522	51,108
c-fat500-5	500	23,191	64	205,864	64	7.81	292,208	224,323	289,644
c-fat500-10	500	46,627	126	804,000	126	3.97	1,174,240	846,003	1,122,240
DSJC500_5	500	62,624	13	9,626	71	7.04	760,032	371,513	445,446
DSJC1000_5	1,000	249,826	15	12,054	122	8.20	2,695,560	1,144,330	1,321,400
gen200_p0.9_44	200	17,910	44	94,362	63	3.17	441,099	262,929	327,415
gen200_p0.9_55	200	17,910	55	150,839	77	2.60	590,043	387,808	467,936
gen400_p0.9_55	400	71,820	55	—	77	5.19	1,107,870	458,318	528,254
gen400_p0.9_65	400	71,820	65	—	102	3.92	1,528,460	780,508	894,911
gen400_p0.9_75	400	71,820	75	—	103	3.88	1,521,530	784,813	910,735
hamming6-2	64	1,824	32	32,736	32	2.00	60,192	33,152	47,120
hamming6-4	64	704	4	396	7	9.14	4,264	1,907	2,145
hamming8-2	256	31,616	128	800,624	130	1.97	1,558,810	945,680	1,277,780
hamming8-4	256	20,864	16	12,360	24	10.67	146,767	42,399	53,533
hamming10-2	1,024	518,656	512	—	537	1.91	26,113,600	15,730,300	21,388,200
hamming10-4	1,024	434,176	—	—	90	11.38	2,798,370	613,716	767,997
johnson8-2-4	28	210	4	192	6	4.67	1,079	470	678
johnson8-4-4	70	1,855	14	6,552	19	3.68	29,569	15,934	19,786
johnson16-2-4	120	5,460	8	3,808	15	8.00	51,783	14,375	20,263
johnson32-2-4	496	107,880	16	—	31	16.00	454,263	79,785	90,891
keller4	171	9,435	11	6,745	33	5.18	143,426	71,782	96,508
keller5	776	225,990	27	—	123	6.31	2,693,370	1,058,210	1,381,980
keller6	3,361	4,619,898	—	—	282	11.92	24,531,900	6,110,990	7,601,260
MANN_a9	45	918	16	5,460	21	2.14	15,019	8,546	14,668
MANN_a27	378	70,551	126	—	144	2.63	2,390,210	1,100,410	1,696,000
MANN_a45	1,035	533,115	345	—	375	2.76	17,904,100	7,256,760	11,639,900

MANN_a81	3,321	5,506,380	1100	—	1160	2.86	184,688,000	69,376,700	111,821,000
p_hat300-1	300	10,933	8	3,321	24	12.50	77,256	42,970	50,724
p_hat300-2	300	21,928	25	31,564	46	6.52	291,593	160,414	187,784
p_hat300-3	300	33,390	36	63,390	72	4.17	719,292	386,513	452,424
p_hat500-1	500	31,569	9	4,764	36	13.89	200,341	97,409	116,933
p_hat500-2	500	62,946	36	63,870	69	7.25	797,997	385,609	430,757
p_hat500-3	500	93,800	50	—	112	4.46	1,901,550	950,409	1,102,070
p_hat700-1	700	60,999	11	5,185	44	15.91	344,530	154,795	176,541
p_hat700-2	700	121,728	44	—	89	7.87	1,399,730	655,765	721,643
p_hat700-3	700	183,010	62	—	148	4.73	3,427,090	1,675,380	1,932,600
p_hat1000-1	1,000	122,253	10	5,436	56	17.86	637,294	260,576	287,901
p_hat1000-2	1,000	244,799	46	—	116	8.62	2,607,780	1,100,310	1,231,560
p_hat1000-3	1,000	371,746	68	—	198	5.05	6,770,190	3,042,040	3,459,520
p_hat1500-1	1,500	284,923	12	7,135	78	19.23	1,377,910	510,199	564,782
p_hat1500-2	1,500	568,960	65	—	169	8.88	5,751,650	2,358,580	2,633,230
p_hat1500-3	1,500	847,244	94	—	284	5.28	14,909,000	6,432,260	7,197,470
san200_0.7_1	200	13,930	30	45,295	31	6.45	191,976	76,677	84,637
san200_0.7_2	200	13,930	18	15,073	18	11.11	122,916	28,141	29,265
san200_0.9_1	200	17,910	70	242,710	74	2.70	582,448	367,724	436,794
san200_0.9_2	200	17,910	60	178,468	75	2.67	578,346	370,206	446,774
san200_0.9_3	200	17,910	44	96,764	56	3.57	430,170	229,653	263,921
san400_0.5_1	400	39,900	13	7,442	14	28.57	129,393	17,229	17,855
san400_0.7_1	400	55,860	40	77,719	41	9.76	544,666	143,893	154,432
san400_0.7_2	400	55,860	30	44,155	30	13.33	413,269	80,724	83,551
san400_0.7_3	400	55,860	22	24,727	29	13.79	338,061	74,233	78,142
san400_0.9_1	400	71,820	100	—	107	3.74	1,766,750	876,509	976,900
san1000	1,000	250,500	15	10,661	16	62.50	376,843	23,456	23,840
sanr200_0.7	200	13,868	18	16,398	51	3.92	298,303	175,001	214,149
sanr200_0.9	200	17,863	42	—	81	2.47	632,063	433,622	509,992
sanr400_0.5	400	39,984	13	8,298	61	6.56	507,447	259,181	319,893
sanr400_0.7	400	55,869	21	22,791	90	4.44	1,082,260	577,693	686,163

In Table 3, the DIMACS instances are instead grouped into families based on the similarity of their names. For each family, the table reports its name and the number of instances it contains (column #). The next columns provide the minimum, maximum, and average *relative ratio* obtained by UB_1 across the instances in each family, computed for each instance and each upper bound as the ratio between its value and that of the bound achieving the minimum value (i.e., $\min\{UB_1, UB_2, UB_3\}$) on the same instance. These are followed by the corresponding statistics for UB_3 . The value of UB_2 is consistently lower than those of UB_1 and UB_3 across the entire testbed of instances. On the sole **c-fat200-1** and **c-fat200-5** instances, UB_2 has a value greater than that of UB_1 . For this reason, Table 3 does not contain columns regarding the relative ratio achieved by UB_2 , which is equal to 1 for all instances but two. Specifically, on the **c-fat200-1** instance, UB_2 achieves a relative ratio of 1.05, while on the **c-fat200-5** instance it equals 1.14.

Table 3: Relative ratio achieved by UB_1 and UB_3 on DIMACS instances, grouped by family. For each bound and family: minimum, maximum, and average relative ratio (ratio between the bound value and $\min\{UB_1, UB_2, UB_3\}$). A DSatur coloring $\mathcal{C}$ is used for computing UB_1 and UB_2 . For UB_2 , the independent sets of $\mathcal{C}$ are sorted in non-decreasing order of cardinality. UB_3 uses $k = |\mathcal{C}|$. Since UB_2 achieves a relative ratio of 1.00 on all instances but two (*c-fat200-1* and *c-fat200-5*), its data are omitted from the table.

Families of instances	#	relative ratio of UB_1			relative ratio of UB_3		
		min	max	avg	min	max	avg
brock	12	1.67	2.16	1.89	1.16	1.26	1.18
C	7	1.45	2.79	1.93	1.13	1.22	1.17
c-fat	7	1.00	1.39	1.19	1.19	1.33	1.26
DSJC	2	2.05	2.36	2.20	1.15	1.20	1.18
gen	5	1.52	2.42	1.90	1.15	1.25	1.18
hamming	6	1.65	4.56	2.56	1.12	1.42	1.30
johnson	4	1.86	5.69	3.36	1.14	1.44	1.31
keller	3	2.00	4.01	2.85	1.24	1.34	1.30
MANN	4	1.76	2.66	2.26	1.54	1.72	1.62
p_hat	15	1.80	2.70	2.17	1.10	1.20	1.14
san	11	1.56	16.07	4.63	1.02	1.21	1.09
sanr	4	1.46	1.96	1.75	1.18	1.23	1.21
Tot/Avg	80	2.44			1.21		

The results reported in Table 3 provide insights on the relative strength of the bounds: it emerges, for example, that UB_1 is the upper bound which tends to have the worst performance. Indeed, its average relative ratio is more than 2.4, and for some families of instances (like *johnson* and *san*) it even reaches values higher than 3. On the other hand, UB_3 seems to ensure a more stable practical performance, being on average 1.2 times larger than the tightest upper bound among the three.

It is also worth noting that computing UB_1 is considerably more time-consuming than UB_2 and UB_3 , as it requires solving Model (3). For instances with many edges, and consequently a large number of variables, solving the model can take several tens of seconds and, in some cases, even exceed a few hundred seconds. In contrast, the evaluation of UB_2 and UB_3 is extremely fast and incurs negligible computational time.

In order to assess the robustness of the upper bounds to the specific coloring used, we also tested each instance with five additional colorings computed via a randomized algorithm. This algorithm first generates a random ordering of the vertices and then assigns each vertex, in that order, to the first available color that does not contain any of its neighbors. We refer to a coloring obtained via this randomized coloring algorithm as *random coloring*. Results show that all the upper bounds generally exhibit weaker performance when computed using random colorings, compared to those obtained with the DSatur coloring. This is likely due to DSatur’s tendency to produce colorings with fewer independent sets. The value of UB_3 is indeed directly related to the value of k , which in our tests coincides with the cardinality of the coloring. As far as the other two bounds are concerned, our computational experiments show that, as the cardinality of coloring increases, the values of both UB_1 and UB_2 tend to increase accordingly.

6.4 Results for RANDOM instances

In this section, we present the results of the tests conducted on the RANDOM instances. Throughout this section, all bounds are computed using a DSatur coloring $\mathcal{C}$. Specifically, UB_1 and UB_2 use $\mathcal{C}$ directly, with the independent sets of $\mathcal{C}$ sorted in non-decreasing order of cardinality for UB_2 ; UB_3 uses instead $k = |\mathcal{C}|$.

To compare the quality of the three bounds independently of the absolute weight scale of each instance, we measure their strength via the *strength ratio*, defined as the ratio between the bound value and the optimal value of the instance. A strength ratio of 1 indicates a bound whose value coincides with the optimal value of the instance, while values greater than 1 indicate overestimation: a bound with a lower strength ratio is therefore stronger. Figure 9 displays the strength ratio of the three bounds across different instance sizes, ranging from 10 to 90 vertices, using a box plot representation. Each box summarizes the distribution of the strength ratio achieved by one of the bounds for all instances of the same size, showing the median, interquartile range, and potential outliers. Specifically, each box plot is based on 90 instances with densities ranging from 10% to 90%, resulting in a total of 90 upper bound evaluations. The results reveal that the median strength ratio for all the bounds is significantly large even for relatively small instances (i.e., for $|V| \geq 30$), confirming the general weakness of these upper bounds, as previously observed on the DIMACS benchmarks. On these instances, the average strength ratio is 3.5 for UB_1 , 2.7 for UB_2 , and 3.2 for UB_3 , with extreme cases exceeding 8 for UB_1 and UB_3 , while the maximum ratio achieved by UB_2 is around 6.6. Moreover, the strength ratio shows a clear upward trend with increasing graph size, highlighting the growing difficulty of obtaining tight upper bounds as $|V|$ increases. Similarly, the variability of the ratio values increases for larger instances, as indicated by the growing interquartile range. This suggests that the performance of the upper bounds, in terms of deviation from $\omega(G, c)$, becomes less consistent as the graph size grows.

Figure 9: Box plot representation of the strength ratio of the three upper bounds on RANDOM instances, grouped by number of vertices from 10 to 90. Box plots for UB_1 are filled with a blue diagonal line pattern, those for UB_2 with a red dotted pattern, and those for UB_3 with a green crossed-line pattern. A DSatur coloring $\mathcal{C}$ is used for computing UB_1 and UB_2 . For UB_2 , the independent sets of $\mathcal{C}$ are sorted in non-decreasing order of cardinality. UB_3 uses $k = |\mathcal{C}|$.

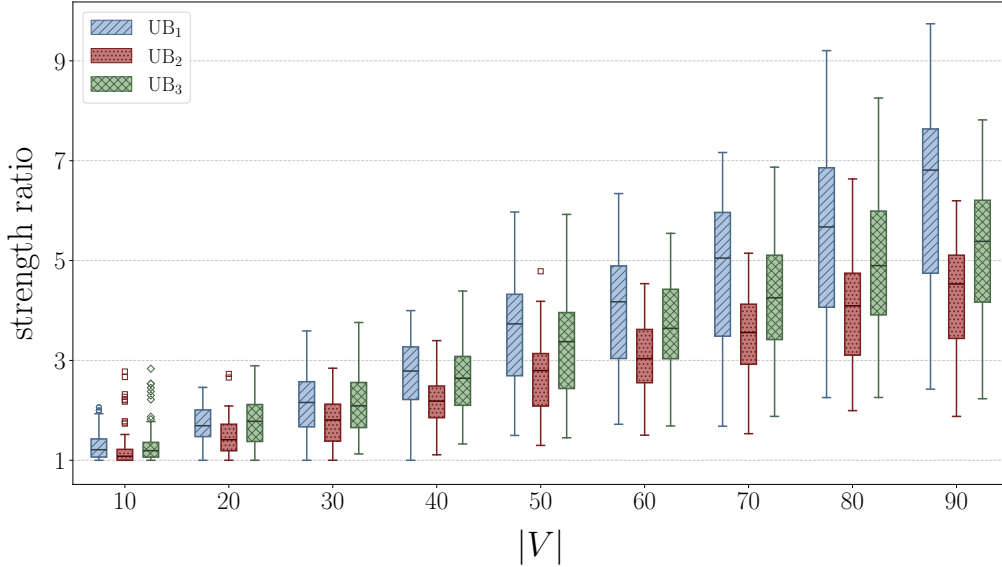
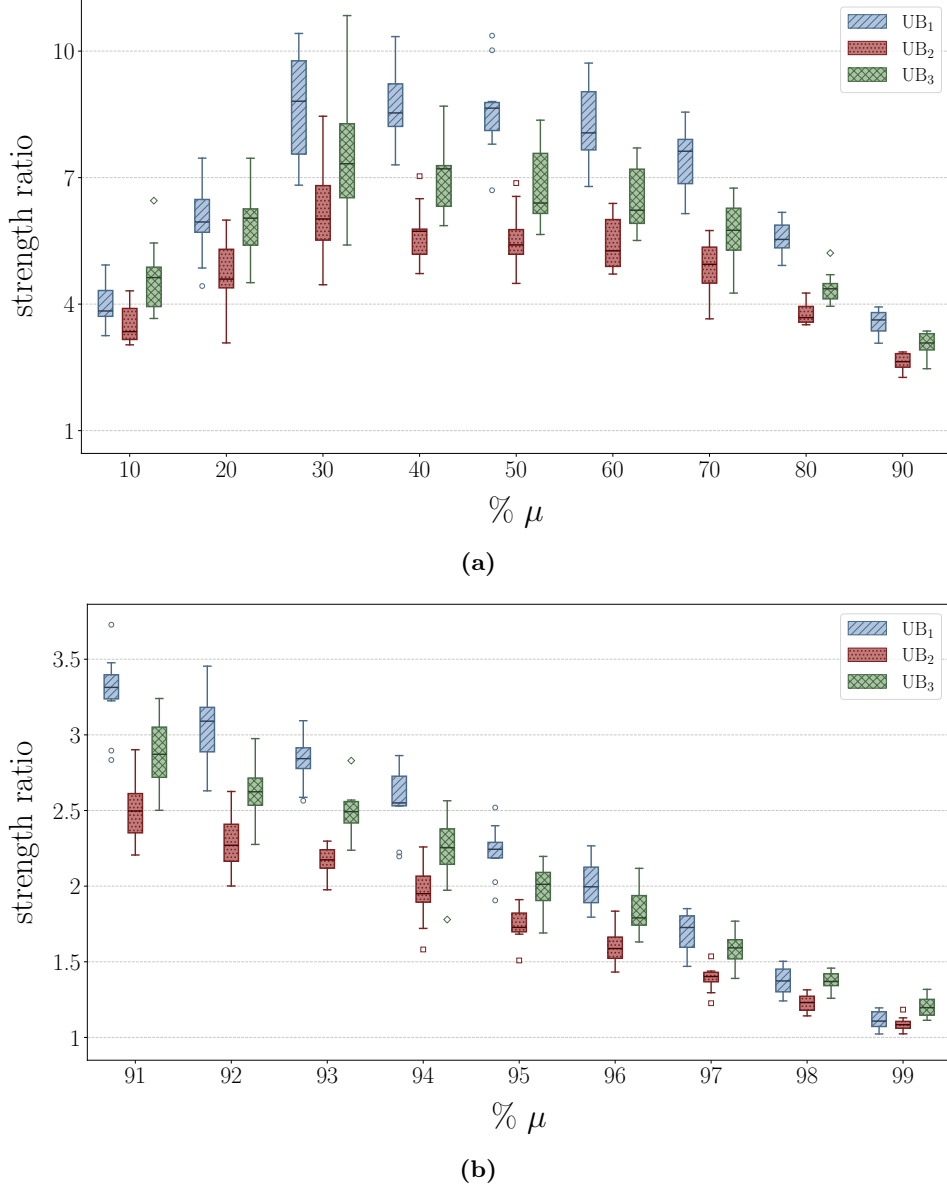


Figure 10 illustrates the influence of graph density on the strength ratio of the three upper bounds, focusing specifically on instances with $|V| = 100$, using a box plot representation. Hence, each group consists of 10 instances. Subfigure (a) shows results for densities from 10% to 90%, while Subfigure (b) focuses on high-density values from 91% to 99%. The results show that the three upper bounds perform worst at low to moderate densities (below 70%). Conversely, for denser graphs, a monotonic decrease in the median strength ratio is observed, with the lowest values—around 1.02—attained for highly dense instances ($\mu = 0.99$). As the box plots reveal, all the upper bounds exhibit weak performance across most density levels for the instances with $|V| = 100$, with considerable median strength ratios for nearly all density values. Notably, sparse graphs appear to be more challenging than dense ones: for 10%-dense instances, both UB_1 and UB_2 have a median ratio of more than 3, while that of UB_3 is approximately 4.6. Finally, and in line with the results observed for the DIMACS instances, the box plots confirm that UB_2 consistently outperforms UB_1 and UB_3 in terms of strength ratio, regardless of the edge density or size of the graph.

Similarly to what has been done for DIMACS instances, we have also tested each RANDOM instance using five random colorings, in addition to the DSatur one. Results confirm that, on average, using DSatur as a coloring algorithm guarantees a better performance than using random colorings, for all three upper bounds. Indeed, the latter tend to produce colorings with one more independent set, on average, than those obtained via DSatur. The average advantage of DSatur is moderate for UB_1 , but grows substantially for UB_2 and for UB_3 , and it is more pronounced on high density instances.

Figure 10: Box plot representation of the strength ratio of the three upper bounds on RANDOM instances with $|V| = 100$, grouped by edge density. Box plots for UB_1 are filled with a blue diagonal line pattern, those for UB_2 with a red dotted pattern, and those for UB_3 with a green crossed-line pattern. A DSatur coloring $\mathcal{C}$ is used for computing UB_1 and UB_2 . For UB_2 , the independent sets of $\mathcal{C}$ are sorted in non-decreasing order of cardinality. UB_3 uses $k = |\mathcal{C}|$.



6.5 Strength of the upper bounds in a branch-and-bound context

The results shown in the previous sections have been obtained computing each upper bound on the full original graph G . However, in the state-of-the-art branch-and-bound algorithms for the EWMCP, the bounds are computed multiple times: at each node of the search tree, a partial clique $C \subset V$ has already been fixed in a partial solution, and the upper bound is computed only on the subgraph $G[L]$ induced by the remaining candidate vertices $L \subset V$, namely the vertices adjacent to all those in C . The total upper bound at that node is then $W(C) + UB(C, L)$, where $W(C)$ denotes the total edge-weight of the current clique, and $UB(C, L)$ is an upper bound on

the maximum contribution that the vertices of L can give to the current partial solution value. The idea for computing $\text{UB}(C, L)$ is common to all the bounds studied in this article: each candidate vertex $v \in L$ is adjacent, by definition, to every vertex currently in C , so it *absorbs* the weights of the edges connecting it to C . Formally, for every $v \in L$, define $\vartheta_v := \sum_{u \in C} c_{\{u, v\}}$ as the total weight that v would contribute to the clique C if added to it. The total upper bound at the current node then decomposes as $W(C) + \text{UB}(C, L)$, where $\text{UB}(C, L)$ is computed on $G[L]$ with ϑ_v playing the role of an additional offset added to each vertex $v \in L$. For UB_2 and UB_3 , this offset is incorporated directly into the vertex weights σ_v and η_v , as described in Shimizu et al. (2020) and Hosseinian et al. (2020), respectively. For UB_1 , it is instead taken into account by increasing the right-hand side of Constraints (3c) by ϑ_v for vertices $v \in L$, and solving the LP on the subgraph $G[L]$ with this modified formulation. To assess how quickly each bound is able to prune nodes in practice, we simulate a single random path through the branching tree. The simulation proceeds as follows. Starting from an empty clique $C = \emptyset$ and candidate set $L = V$, at each step a random vertex v is selected from L and added to C ; then L is updated to $L \cap N(v)$, where $N(v)$ denotes the neighborhood of v , thus retaining only the vertices adjacent to all current members of C . After each update, $G[L]$ is re-colored using the DSatur heuristic. The independent sets of the resulting coloring are sorted in non-decreasing order of cardinality; for UB_3 , we set $k = |\mathcal{C}|$, where $\mathcal{C}$ denotes the coloring computed at the current node. The simulation terminates either when $W(C) + \text{UB}(C, L) \leq \omega(G, \mathbf{c})$ (the node is pruned by the bound) or when L becomes empty (a maximal clique has been reached without pruning). We refer to the number of vertices in C at termination as the *pruning depth*. Observe that we use the optimal EWMCP value $\omega(G, \mathbf{c})$ as the incumbent value throughout the simulation. This constitutes an idealized setting: in a real branch-and-bound algorithm, the incumbent is a lower bound to $\omega(G, \mathbf{c})$, and its value may significantly affect the depth reached before being able to prune a node. Consequently, this experiment is designed solely to compare the relative node-dependent pruning depth of the bounds under a common idealized protocol; it does not evaluate overall branch-and-bound running time or total search-tree size. The simulation is run on the 48 DIMACS instances for which $\omega(G, \mathbf{c})$ is known, exploring 10 independent random branchings per instance, yielding 480 runs per bound.

Table 4 reports the minimum, maximum and average pruning depth over the 10 seeds for each instance and each upper bound. Under this protocol, UB_2 achieves the lowest average depth on 42 of the 48 instances. Such dominance is confirmed by the average depth values: UB_2 prunes at a shallower depth (4.1 on average) than both UB_1 and UB_3 (4.9 and 4.8, respectively). For some instances, the performance gap between UB_2 and the other two bounds is particularly relevant: as an example, for the **hamming8** – 2 instance the average pruning depth obtained using UB_2 as an upper bound is approximately half of those obtained with UB_1 and UB_3 . Conversely, on all the instances belonging to the **c-fat** family, UB_1 proves to be the strongest upper bound: indeed, when using UB_1 as an upper bound, pruning occurs at depth 1 in all the branching simulations performed on these instances. These results concern relative node-dependent pruning depth only and should not be interpreted as evidence about the running time or total search-tree size of a complete branch-and-bound algorithm.

Table 4: Minimum, maximum, and average pruning depth over 10 simulated branchings, for each of the 48 DIMACS instances with known optimum. At each node of the simulation, all bounds are recomputed on the current subgraph using a DSatur coloring $\mathcal{C}$. For UB_2 , the independent sets of each coloring are sorted in non-decreasing order of cardinality. UB_3 uses $k = |\mathcal{C}|$.

Instance	pruning depth with UB_1			pruning depth with UB_2			pruning depth with UB_3		
	min	max	avg	min	max	avg	min	max	avg
brock200_1	5	8	6.8	5	7	6.4	5	8	7.0
brock200_2	3	4	3.4	3	4	3.2	3	4	3.4
brock200_3	4	5	4.7	4	5	4.5	5	6	5.1
brock200_4	4	6	4.8	3	6	4.7	4	6	4.9
brock400_3	6	7	6.9	6	7	6.7	6	7	6.9
brock400_4	6	8	7.1	6	7	6.5	6	8	7.0
C125.9	7	11	9.3	6	11	8.6	7	13	9.6
c-fat200-1	1	1	1.0	1	1	1.0	1	4	1.4
c-fat200-2	1	1	1.0	1	1	1.0	1	4	2.4
c-fat200-5	1	1	1.0	1	2	1.2	2	4	2.2
c-fat500-1	1	1	1.0	1	1	1.0	1	1	1.0
c-fat500-2	1	1	1.0	1	3	1.2	1	3	1.2
c-fat500-5	1	1	1.0	1	4	1.3	2	5	2.7
c-fat500-10	1	1	1.0	1	4	1.4	2	4	2.7
DSJC500_5	4	5	4.7	4	5	4.3	4	5	4.7
DSJC1000_5	5	6	5.2	5	5	5.0	5	6	5.1
gen200_p0.9_44	9	14	11.4	8	13	10.1	9	14	11.1
gen200_p0.9_55	6	8	7.3	5	7	6.1	6	8	7.3
hamming6-2	3	6	3.9	2	6	2.5	2	6	3.5
hamming6-4	2	3	2.3	2	3	2.9	2	3	2.9
hamming8-2	11	16	12.1	5	12	6.2	10	16	11.7
hamming8-4	3	5	3.9	3	5	3.6	3	5	4.0
johnson8-2-4	2	2	2.0	2	3	2.6	2	3	2.6
johnson8-4-4	3	5	3.8	3	7	3.5	3	7	4.0
johnson16-2-4	6	6	6.0	6	6	6.0	6	6	6.0
keller4	4	5	4.2	3	4	3.7	3	5	4.0
MANN_a9	9	12	10.1	5	12	8.7	9	15	11.6
p_hat300-1	2	3	2.3	2	3	2.3	2	4	2.5
p_hat300-2	2	7	4.3	2	7	3.7	2	8	4.3
p_hat300-3	5	9	6.4	4	7	5.5	5	7	6.1
p_hat500-1	2	4	2.5	2	3	2.4	2	4	2.5
p_hat500-2	2	8	4.3	2	6	3.7	2	8	4.1
p_hat700-1	2	4	3.1	2	4	3.1	3	4	3.3
p_hat1000-1	3	4	3.2	3	4	3.2	3	4	3.3
p_hat1500-1	3	4	3.3	3	4	3.2	3	4	3.5
san200_0.7_1	3	5	3.5	2	3	2.6	3	4	3.2
san200_0.7_2	3	8	6.5	2	5	3.8	3	6	4.2
san200_0.9_1	4	8	5.9	3	5	4.0	4	7	5.5
san200_0.9_2	6	8	7.1	5	6	5.7	6	8	6.7
san200_0.9_3	9	11	10.3	7	10	8.7	9	11	10.2
san400_0.5_1	3	5	4.0	2	2	2.0	2	3	2.3

san400_0.7_1	3	9	5.5	3	3	3.0	3	4	3.9
san400_0.7_2	5	8	6.3	3	5	3.8	3	5	4.7
san400_0.7_3	6	8	6.8	3	6	4.4	4	7	5.1
san1000	5	6	5.1	2	2	2.0	2	3	2.8
sanr200_0.7	5	6	5.5	5	6	5.3	5	6	5.6
sanr400_0.5	4	5	4.1	4	5	4.1	4	5	4.1
sanr400_0.7	6	8	7.1	6	8	7.0	6	8	7.2
Avg			4.9			4.1			4.8

6.6 A comparative analysis of LP-based upper bounds

In this subsection, we compare five LP-based upper bounds to $\omega(G, \mathbf{c})$. The first three are variants of the bound $\text{UB}_1(G, \mathbf{c}, \tilde{\mathcal{I}})$, differing in the family $\tilde{\mathcal{I}}$ of independent sets used in Model (3). The remaining two are obtained by solving the linear relaxation of two ILP formulations for the EWMCP.

The optimal value $\text{UB}_1(G, \mathbf{c}, \tilde{\mathcal{I}})$ of Model (3) depends on the family $\tilde{\mathcal{I}}$ of independent sets over which variables $\boldsymbol{\pi}$ are defined. As introduced in Section 2.1, San Segundo et al. (2019) restrict this family to a coloring $\mathcal{C}$ (i.e., a partition) of the vertex set V . It is therefore natural to ask whether enlarging $\tilde{\mathcal{I}}$ yields a tighter bound, and at what computational cost. We consider two extensions of the baseline bound $\text{UB}_1 = \text{UB}_1(G, \mathbf{c}, \mathcal{C})$, here defined over a single DSatur coloring. The first, denoted $\text{UB}_1^\cup$, takes $\tilde{\mathcal{I}}$ to be the union of the six colorings (one obtained via DSatur and 5 random colorings) described in Section 6.3, thus allowing vertices to appear in more than one independent set. The second, denoted UB_1^* , sets $\tilde{\mathcal{I}} = \mathcal{I}$, i.e., the collection of all independent sets of G . Since every independent set is contained in a maximal independent set, it is sufficient computationally to consider only maximal independent sets. In this case, the value of $\text{UB}_1(G, \mathbf{c}, \mathcal{I})$ coincides with the optimal value of the linear relaxation of Model (2). The computation of $\text{UB}_1(G, \mathbf{c}, \mathcal{I})$ requires solving a formulation with an exponential number of variables, which we handle via column generation. The corresponding pricing subproblems are Maximum Weighted Independent Set problems with suitably defined vertex weights, which we solve by finding a Maximum Vertex Weighted Clique on the complement graph of G , using the TSM algorithm (Jiang et al., 2018).

We place these variants within a broader landscape of LP-based bounds by introducing two further bounds derived from solving the linear relaxation of two ILP formulations for the EWMCP. The first formulation is denoted as F1, and it is the natural ILP formulation of the problem, adapted from the one originally proposed by Park et al. (1996). This formulation corresponds to Model (2), where Constraints (2e) are defined over the set of all non-edges of G , i.e., over $\bar{E} = \{\{u, v\} : \{u, v\} \notin E\}$. Notice that the dual of the linear relaxation of F1 coincides with Model (3), having $\tilde{\mathcal{I}} = \bar{E}$.

The second formulation, detailed in Gouveia and Martins (2015), is denoted as F11. It is obtained from F1 by adding the following two classes of valid inequalities to the model:

$$\sum_{e \in \delta(u)} y_e \leq (k-1) x_u, \quad u \in V, \quad \text{and} \quad \sum_{u \in V} x_u \leq k, \quad (18)$$

where k is any upper bound on the clique number of G . The first family of inequalities ensures that an edge can be selected only if its endpoints are both in the clique, and limits the number of selected edges incident to each node to at most $k-1$. The second constraint imposes instead a global upper bound of k on the number of selected vertices. Gouveia and Martins (2015) showed that the linear relaxation of F11 is not dominated by that of any other compact ILP formulation for the EWMCP they consider, in the sense that neither provides a tighter bound on every instance.

The upper bounds obtained by solving the linear relaxations of formulations F1 and F11 are henceforth referred to as UB_{F1} and UB_{F11} , respectively. In the following computational experiments, the value of k for Constraints (18) is set to the cardinality of the coloring obtained via the DSatur algorithm, for all the tested instances.

Table 5 reports the performance of all five upper bounds on 43 DIMACS instances with known optimal values. Instances are grouped by family; for each family, the table reports the number of instances, the average relative ratio (the ratio of each bound’s value to the minimum one across all five bounds on the same instance), and the average time ratio (the ratio of each bound’s computing time to the minimum one across all five bounds on the same instance). A value of 1.00 in either of these ratios indicates that the corresponding bound achieves the best value, respectively the shortest computing time, among the five upper bounds on that instance. Four instances of the **p_hat** family, as well as the DSJC1000_5 instance, are excluded from the analysis since the column generation procedure for UB_1^* did not converge within a 20-hour time limit.

The results reveal a clear hierarchy. UB_{F1} is the weakest of the five upper bounds by a wide margin: its average relative ratio of 22.62 reflects that, on most instances, it is far from the tightest among the 5 LP-based upper bounds considered. This is most pronounced on the **san** family, where UB_{F1} is on average more than 77 times larger than the best bound, confirming that restricting $\mathcal{J}$ to pairs is a poor choice in practice. Its computing time is comparable to that of UB_1 (average time ratio 2.65), so the bound weakness is not offset by any computational advantage. Within the $\text{UB}_1(G, \mathbf{c}, \mathcal{J})$ family, enlarging $\mathcal{J}$ yields a diminishing improvement in bound tightness. Moving from UB_1 (average relative ratio 2.31) to $\text{UB}_1^\cup$ (2.17) slightly reduces the gap, at a moderate computational cost of 2.64 times in terms of computing time. Extending to UB_1^* (having an average relative ratio of 2.02) achieves a further modest improvement, but with an average time ratio of around 1330 – dominated by the **p_hat** family, where column generation requires on average over 13000 times the minimum computing time. The gain in bound tightness from $\text{UB}_1^\cup$ to UB_1^* is therefore not justified by the additional computational effort on almost all the instance families. The bound UB_{F11} stands apart from all the other LP-based bounds considered above. With an average relative ratio of 1.02, it is the tightest bound on basically every instance, achieving an average ratio of exactly 1.00 on eight of the twelve families. It reaches this quality at an average time ratio of 10.77, substantially less

than UB_1^* and comparable to $UB_1^\cup$ on most families, with the exception of **c-fat**, **hamming**, **johnson** and **MANN**, where its time ratio reaches high values. The only family where UB_{F11} shows a significant performance gap with respect to the best bound is **keller**, where it achieves a relative ratio of 1.28, while UB_1^* reaches 1.00.

We observe that three variants of $UB_1(G, \mathbf{c}, \tilde{\mathcal{J}})$ (UB_1 , $UB_1^\cup$ and UB_1^*) differ substantially in the number of independent sets they employ (i.e., on the value of $|\tilde{\mathcal{J}}|$): on the 43 considered instances, UB_1 uses on average 48.3 independent sets, corresponding to a single DSatur coloring. $UB_1^\cup$ enlarges this pool adding the independent sets from the five random colorings and reaching an average of 314.9 sets (approximately, a 6x increase that is stable across families). UB_1^* , by contrast, requires substantially fewer independent sets: on most families the column generation procedure adds only a moderate number of sets, yielding an average of 133.9, well below $UB_1^\cup$. The main exception is the **p_hat** family, where UB_1^* reaches an average of 482.5 sets, surpassing $UB_1^\cup$ (369.2), and where the column generation procedure did not terminate within the time limit on four of the eight instances. The high computational cost of UB_1^* is therefore not due to a large number of generated columns – on most instances, the column generation procedure terminates after adding relatively few independent sets – but rather to the cost of solving the pricing subproblem at each iteration.

Two main conclusions follow from these results. First, the gap between UB_1 and stronger bounds within the $UB_1(G, \mathbf{c}, \tilde{\mathcal{J}})$ family seems not to be closed by enlarging $\tilde{\mathcal{J}}$ alone: even UB_1^* , defined over the full independent set family $\mathcal{J}$, retains an average relative ratio of 2.02 at an impractical computational cost. Second, the valid inequalities of F11 significantly strengthen the associated bound: UB_{F11} is tighter than UB_1^* at a fraction of its computing time. Nevertheless, we observe that even UB_{F11} , the strongest LP-based bound considered here, is dominated by UB_2 on 41 out of 43 instances. On the two exceptions (**c-fat500-1** and **hamming6-2**), the difference between the two bounds is negligible (below 2.5%). On average, the LP-based bounds examined in this subsection are therefore both slower and weaker than UB_2 .

Table 5: Performance of the five LP-based upper bounds on 43 DIMACS instances with known optimum, grouped by family. UB_1 uses a single DSatur coloring $\mathcal{C}$; UB_1^U uses the union of six colorings (one DSatur and five random); UB_1^* uses the family $\mathcal{I}$ of all independent sets of G . For UB_{F11} , the parameter k in Constraints (18) is set to $|\mathcal{C}|$, where $\mathcal{C}$ is the DSatur coloring. For each family, the table reports the number of instances, along with the average relative ratio and the average time ratio of each bound.

Instances		UB_1		UB_1^U		UB_1^*		UB_{F1}		UB_{F11}	
		ratio		ratio		ratio		ratio		ratio	
family	#	relative	time	relative	time	relative	time	relative	time	relative	time
brock	6	1.51	1.00	1.33	3.39	1.07	166.85	3.96	2.10	1.01	4.43
C	1	1.42	1.00	1.26	2.29	1.21	43.89	1.96	2.01	1.00	3.40
c-fat	7	1.10	1.00	1.08	2.65	1.08	30.30	6.97	6.29	1.07	20.35
DSJC	*1	1.80	1.08	1.63	2.37	1.20	1069.34	7.48	1.00	1.00	2.74
gen	2	1.34	1.00	1.19	2.80	1.08	41.40	2.40	2.32	1.01	5.36
hamming	4	2.05	1.00	1.91	2.23	1.90	14.94	8.63	2.07	1.00	19.81
johnson	3	1.97	1.00	1.68	1.77	1.65	47.93	8.06	1.60	1.00	15.77
keller	1	2.04	1.00	1.17	2.71	1.00	43.90	6.87	1.89	1.28	4.90
MANN	1	1.43	1.00	1.39	1.92	1.39	52.03	2.06	1.77	1.00	37.22
p_hat	*4	1.79	1.00	1.62	2.59	1.15	13296.40	7.64	1.77	1.00	12.90
san	10	4.71	1.00	4.66	2.49	4.55	19.80	77.36	1.99	1.00	4.44
sanr	3	1.56	1.00	1.37	3.42	1.08	377.32	4.65	2.01	1.00	4.15
Tot/Avg	43	2.31	1.00	2.17	2.64	2.02	1330.80	22.62	2.65	1.02	10.77

subset of instances with known UB_1^

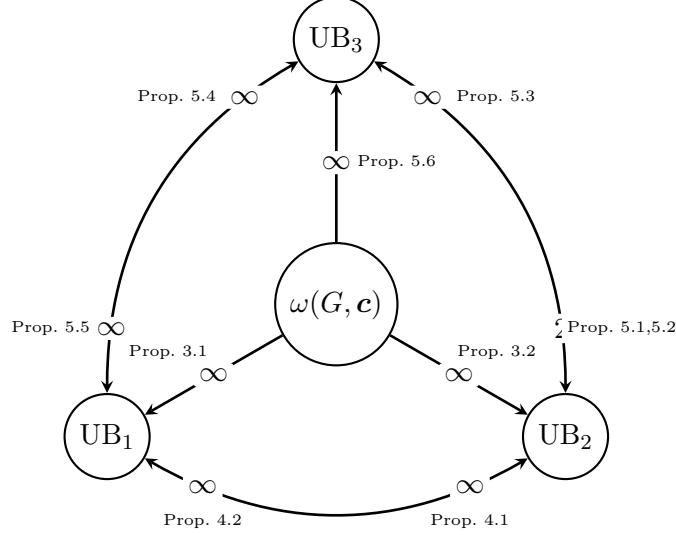
For completeness, we also considered the best value of UB_1 obtained across the DSatur coloring and the five random colorings. This best-over-six strategy provides only a limited improvement over the single-coloring baseline: its average relative ratio decreases from 2.31 to 2.28, with a tighter bound on just 9 of the 43 instances (in the remaining 34 cases, the tightest bound is achieved by DSatur). This confirms the observation made in the previous sections that DSatur is generally more effective than the random colorings. In contrast, pooling the independent sets from all six colorings yields a more substantial improvement, reducing the average relative ratio to 2.17 and producing a bound that is tighter than the best-over-six value on 38 instances (and equal on the remaining 5). The complete best-over-six results are available in the online computational material.

7 Conclusions

In this study, we conducted the first theoretical and computational comparison of the three main upper bounds on $\omega(G, \mathbf{c})$, the optimal value of the EWMCP, proposed independently by San Segundo et al. (2019), Shimizu et al. (2020), and Hosseinian et al. (2020). Our theoretical analysis shows that none of the three upper bounds admits a performance guarantee: for each, there exist instances where its ratio to the optimal EWMCP value is unbounded. Moreover, for every choice of a ratio between two of the three upper bounds, we identified families of instances

for which such ratio achieves the largest possible value. A summary of all the results can be found in Figure 11.

Figure 11: Summary of the theoretical relationships among the three upper bounds and the edge-weighted clique number $\omega(G, c)$, all evaluated with respect to a given coloring $\mathcal{C}$ (with $k = |\mathcal{C}|$ for UB_3). An arrow from $\omega(G, c)$ to an upper bound is labeled with the largest possible value of the ratio of the upper bound to $\omega(G, c)$. An arrow is drawn from an upper bound to the other if the former can dominate the latter. Moreover, the label near the head of the arrow reports the largest possible ratio of the head to the tail. The proposition proving each result is referenced next to the corresponding label.



In particular, the bound proposed by Shimizu et al. (2020) appears to perform better on instances where large independent sets are connected by a few heavy edges, while the one introduced by San Segundo et al. (2019) tends to yield tighter values when the coloring includes a larger number of independent sets. The third bound of Hosseinian et al. (2020) shows qualitative characteristics similar to those of Shimizu et al. (2020), and can improve it by a factor of at most 2, in certain instances. Still, it can be arbitrarily worse than the other two bounds in specific families of instances. These observations were corroborated by our empirical analysis, showing that the first two upper bounds behave differently depending on the graph’s structure. While they perform similarly on small or sparse instances, the upper bound by Shimizu et al. (2020) proves tighter on large, moderately dense graphs, especially around 50% density. In the experiments, the third upper bound tends to perform middle of the road between the other two. It is tighter than the one of San Segundo et al. (2019) for intermediate densities, while it is looser in very sparse and very dense instances. This pattern emerges consistently across the randomly generated graphs, and similar behavior is also observed on the DIMACS benchmark set, highlighting the importance of accounting for graph size and density when selecting or designing bounding strategies for the EWMCP.

An additional practical outcome of our study is that the quality of UB_2 can be significantly improved by appropriately ordering the independent sets of the coloring, and sorting them in non-decreasing order of cardinality turns out to be the best overall strategy on the DIMACS benchmark.

These findings open several lines for future research, including the design of hybrid upper bounds

that leverage the complementary strengths of the three methods based on instance characteristics, as well as their effective integration into exact algorithms such as branch-and-bound frameworks.

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